Calculus

1. L'Hôpital's Rule
2. Extreme Values
3. Concavity and Inflections, Sketching the Graph of a Function
4. Extreme-Value Problems
5. Linear Approximations, Taylor Polynomials
6. Ideas of Integration
7. Method of Substitution
8. Integration by Parts
9. Integrals of Rational Functions
10. Inverse Substitutions
11. Other Methods for Evaluating Integrals

L'Hôpital's Rule

Table 1: Typical Intermediate Forms of Limit

Form	Example

We will focus on $\frac{0}{0}, \frac{\infty}{\infty}$; Other forms can sometimes be transformed to these two

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1^∞	$\lim\limits_{x\to \infty} \left(1 + \frac{1}{x}\right)^x$

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Let f and g be real-valued and differentiable with

$$g' \neq 0 \tag{Denominator$' \neq 0$}$$

on (a, b), $a < b$, $a, b \in \mathbb{R}^\sharp$. Suppose

$$\left(\frac{0}{0}\right)$$

or

$$(\text{Denominator} = \infty)$$

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$$\lim_{x \downarrow a} f(x) = \lim_{x \downarrow a} g(x) = 0 \tag{$\frac{0}{0}$}$$

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If

$$\lim_{x \downarrow a} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}^\sharp$$

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The result holds if $x \downarrow a$ is replaced by $x \uparrow b$ and/or ∞ is replaced by $-\infty$.

L'Hôpital's Rule: Example I

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(Denominator' $\neq 0$) $(x)' = 1 \neq 0$ on $(0, b)$, $b > 0$.

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$$\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{(\sin x)'}{(x)'} = \lim_{x\to 0} \frac{\cos x}{1} = 1$$

L'Hôpital's Rule: Example II

Find

$$\lim_{x\to\infty} \frac{\ln x}{x^{\alpha}}, \quad \alpha > 0$$

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We know from previous class $\lim_{x\to\infty} \frac{\ln x}{x^\alpha} = 0$ by using $\ln x \leqslant x - 1$, $\forall x > 0$.

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Hence

$$\lim_{x\to\infty} \frac{\ln x}{x^\alpha} = \lim_{x\to\infty} \frac{(\ln x)'}{(x^\alpha)'} = \lim_{x\to\infty} \frac{\frac{1}{x}}{\alpha x^{\alpha-1}} = \lim_{x\to\infty} \frac{1}{\alpha x^\alpha} = 0$$

L'Hôpital's Rule: Example III

Find

$$\lim_{x \to 1+} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$$

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$$\lim_{x\to 1+}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)=\lim_{x\to 1+}\frac{x\ln x-(x-1)}{(x-1)\ln x}\qquad\text{(In the form }\frac{0}{0}\text{)}$$

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(Denominator' $\neq 0$) $((x-1)\ln x)'=1-\frac{1}{x}+\ln x\neq 0$ on $(1,b)$, $b>1$.

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Hence

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Hence

$$\lim_{x\to 1+}\frac{x\ln x-(x-1)}{(x-1)\ln x}=\lim_{x\to 1+}\frac{(x\ln x-(x-1))'}{((x-1)\ln x)'}=\lim_{x\to 1+}\frac{1+\ln x-1}{1-\frac{1}{x}+\ln x}=\lim_{x\to 1+}\frac{\ln x}{1-\frac{1}{x}+\ln x}$$

Using L'Hôpital's rule again with

$$\lim_{x \to 1+} \frac{\ln x}{1 - \frac{1}{x} + \ln x}$$

L'Hôpital's Rule: Example III (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x \to 1+} \frac{\ln x}{1 - \frac{1}{x} + \ln x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

L'Hôpital's Rule: Example III (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x\to 1+} \frac{\ln x}{1 - \frac{1}{x} + \ln x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $\left(1 - \frac{1}{x} + \ln x\right)' = \frac{1+x}{x^2} \neq 0$ on $(1, b)$, $b > 1$.

Using L'Hôpital's rule again with

$$\lim_{x\to 1+} \frac{\ln x}{1 - \frac{1}{x} + \ln x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

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Using L'Hôpital's rule again with

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Using L'Hôpital's rule again with

$$\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x}$$

Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

(Denominator' $\neq 0$) $\left(1-\frac{1}{x}+\ln x\right)' = \frac{1+x}{x^2} \neq 0$ on $(1,b)$, $b>1$.

($\frac{0}{0}$) $\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x}$ is of $\frac{0}{0}$ form: $\lim_{x\to 1+} \ln x = \lim_{x\to 1+} 1-\frac{1}{x}+\ln x = 0$

Hence

$$\lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \lim_{x\to 1+} \frac{(\ln x)'}{\left(1-\frac{1}{x}+\ln x\right)'} = \lim_{x\to 1+} \frac{\frac{1}{x}}{\frac{1+x}{x^2}} = \frac{1}{2}$$

So

$$\lim_{x\to 1+} \left(\frac{x}{x-1}-\frac{1}{\ln x}\right) = \lim_{x\to 1+} \frac{x\ln x-(x-1)}{(x-1)\ln x} = \lim_{x\to 1+} \frac{\ln x}{1-\frac{1}{x}+\ln x} = \frac{1}{2}$$

L'Hôpital's Rule: Example IV

Find

$$\lim_{x \to 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

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$$\lim_{x\to 0}\frac{1}{x^2}\ln\frac{\sin x}{x} = \lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2} \qquad (\text{In the form } \frac{0}{0})$$

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Now using L'Hôpital's rule by checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

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(Denominator' $\neq 0$) $(x^2)' = 2x \neq 0$ on $(0, b)$, $b > 0$.

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Hence

$$\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2} = \lim_{x\to 0}\frac{\left(\ln\frac{\sin x}{x}\right)'}{(x^2)'}$$

Find

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}}$$

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Hence

$$\lim_{x\to 0}\frac{\ln\frac{\sin x}{x}}{x^2} = \lim_{x\to 0}\frac{\left(\ln\frac{\sin x}{x}\right)'}{(x^2)'} = \lim_{x\to 0}\frac{\frac{x}{\sin x}\frac{(x\cos x-\sin x)}{x^2}}{2x} = \lim_{x\to 0}\frac{x\cos x-\sin x}{2x^2\sin x}$$

Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{x \cos x - \sin x}{2x^2 \sin x}$$

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Checking the conditions (Denominator' $\neq 0$) and ($\frac{0}{0}$):

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x^2 \sin x)' = 2x^2 \cos x + 4x \sin x \neq 0$ on $(0, b)$, $b > 0$ is small.

Using L'Hôpital's rule again with

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}$$

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Hence

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Hence

$$\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x} = \lim_{x\to 0} \frac{(x\cos x - \sin x)'}{(2x^2 \sin x)'} = \lim_{x\to 0} \frac{-x\sin x}{2x^2 \cos x + 4x \sin x} = \lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}$$

Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{-\sin x}{2x \cos x + 4 \sin x}$$

L'Hôpital's Rule: Example IV (Cont'd)

Using L'Hôpital's rule again with

$$\lim_{x \to 0} \frac{-\sin x}{2x \cos x + 4 \sin x}$$

Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

Using L'Hôpital's rule again with

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Checking the conditions (Denominator' $\neq 0$) and $(\frac{0}{0})$:

(Denominator' $\neq 0$) $(2x\cos x + 4\sin x)' = -2x\sin x + 6\cos x \neq 0$ on $(0, b)$, $b > 0$ is small.

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Hence

$$\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x} = \lim_{x\to 0} \frac{(-\sin x)'}{(2x\cos x + 4\sin x)'} = \lim_{x\to 0} \frac{-\cos x}{-2x\sin x + 6\cos x} = -\frac{1}{6}$$

Using L'Hôpital's rule again with

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So

$$\begin{aligned}\lim_{x\to 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} &= \exp\left\{\lim_{x\to 0} \frac{1}{x^2} \ln \frac{\sin x}{x}\right\} \\ &= \exp\left\{\lim_{x\to 0} \frac{x\cos x - \sin x}{2x^2 \sin x}\right\} = \exp\left\{\lim_{x\to 0} \frac{-\sin x}{2x\cos x + 4\sin x}\right\} = e^{-\frac{1}{6}}\end{aligned}$$

(4.3.29)

$$\lim_{x\to 0} (\cos 2x)^{\frac{1}{x^2}}$$

(4.3.30)

$$\lim_{x\to 0+} \frac{\csc x}{\ln x}$$

(4.3.31)

$$\lim_{x\to 0} \frac{\ln \sin \pi x}{\csc \pi x}$$

(4.3.32)

$$\lim_{x\to 0} (1 + \tan x)^{\frac{1}{x}}$$

(4.3.33) For twice differentiable f, find

$$\lim_{h\to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

(4.3.34) If f has continuous third derivative, find

$$\lim_{h\to 0} \frac{f(x+3h) - 3f(x+h) + 3f(x-h) - f(x-3h)}{h^3}$$

Extreme Values

Let $f : I \mapsto \mathbb{R}$.

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f has an absolute maximum value $f(x_0)$ at $x_0 \in I$ if

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Examples:

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$ has absolute maximum 1 and absolute minimum -1

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$ has absolute maximum 1 and absolute minimum -1

$f(x) = \frac{1}{x}$ on $I = \mathbb{R}$

Let $f : I \mapsto \mathbb{R}$.

f has an absolute maximum value $f(x_0)$ at $x_0 \in I$ if $f(x) \leqslant f(x_0) \quad \forall x \in I$

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Examples:

$f(x) = \sin x$ on $I = \mathbb{R}$ has absolute maximum 1 and absolute minimum -1

$f(x) = \frac{1}{x}$ on $I = \mathbb{R}$ has no absolute maximum and minimum

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The maximum is $f(-1) = 7$, the minimum is $f(2) = -20$.

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Applications

Concavity and Inflections, Sketching the Graph of a Function

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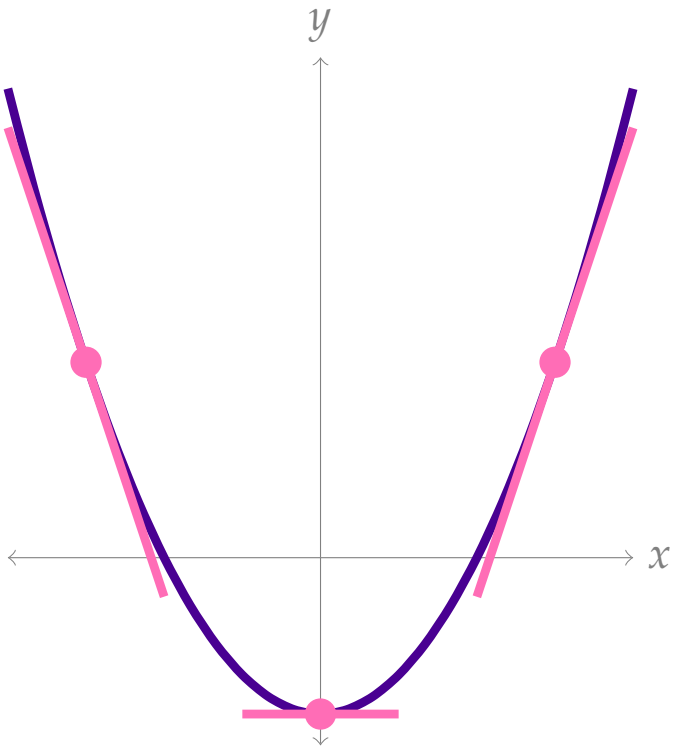
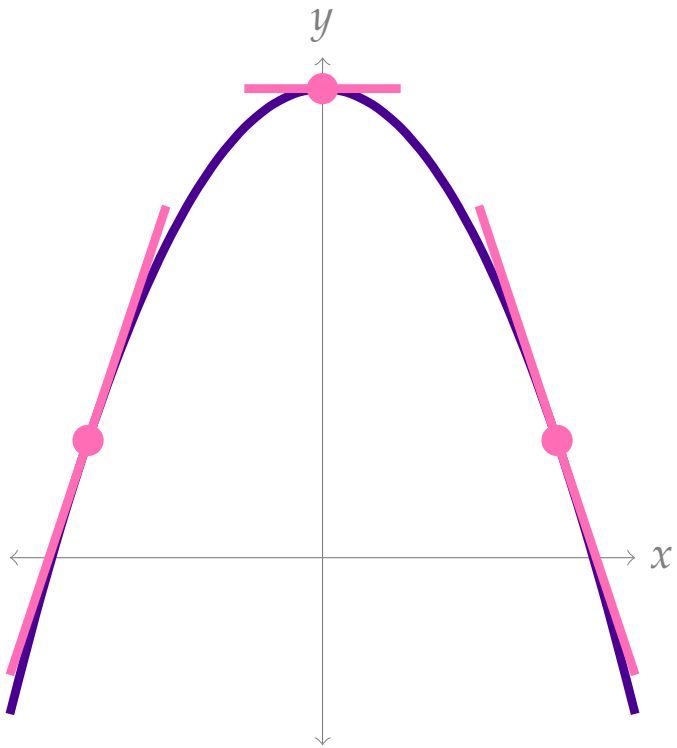


Slopes are increasing

$f''(x) > 0$

"concave up"

tangent line below curve



Slopes are decreasing

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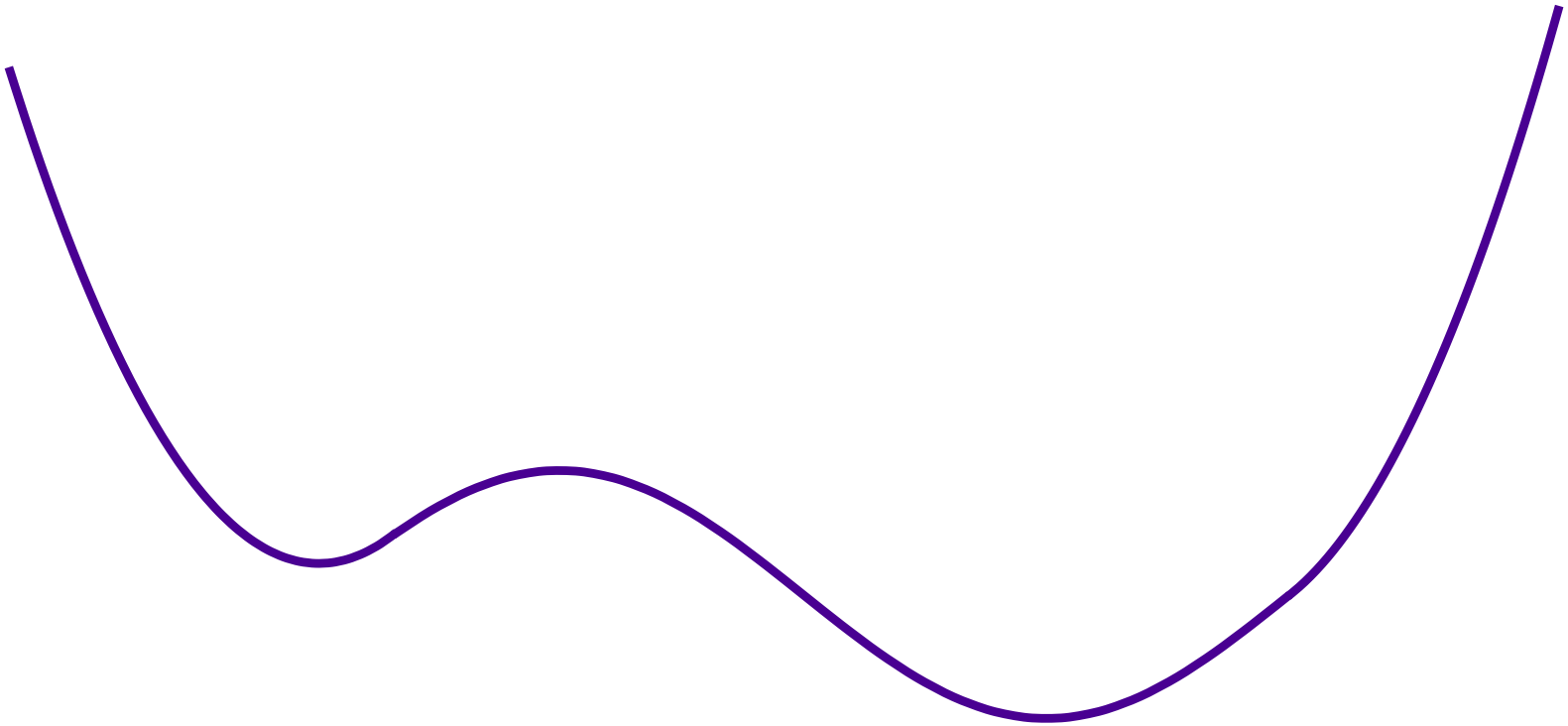
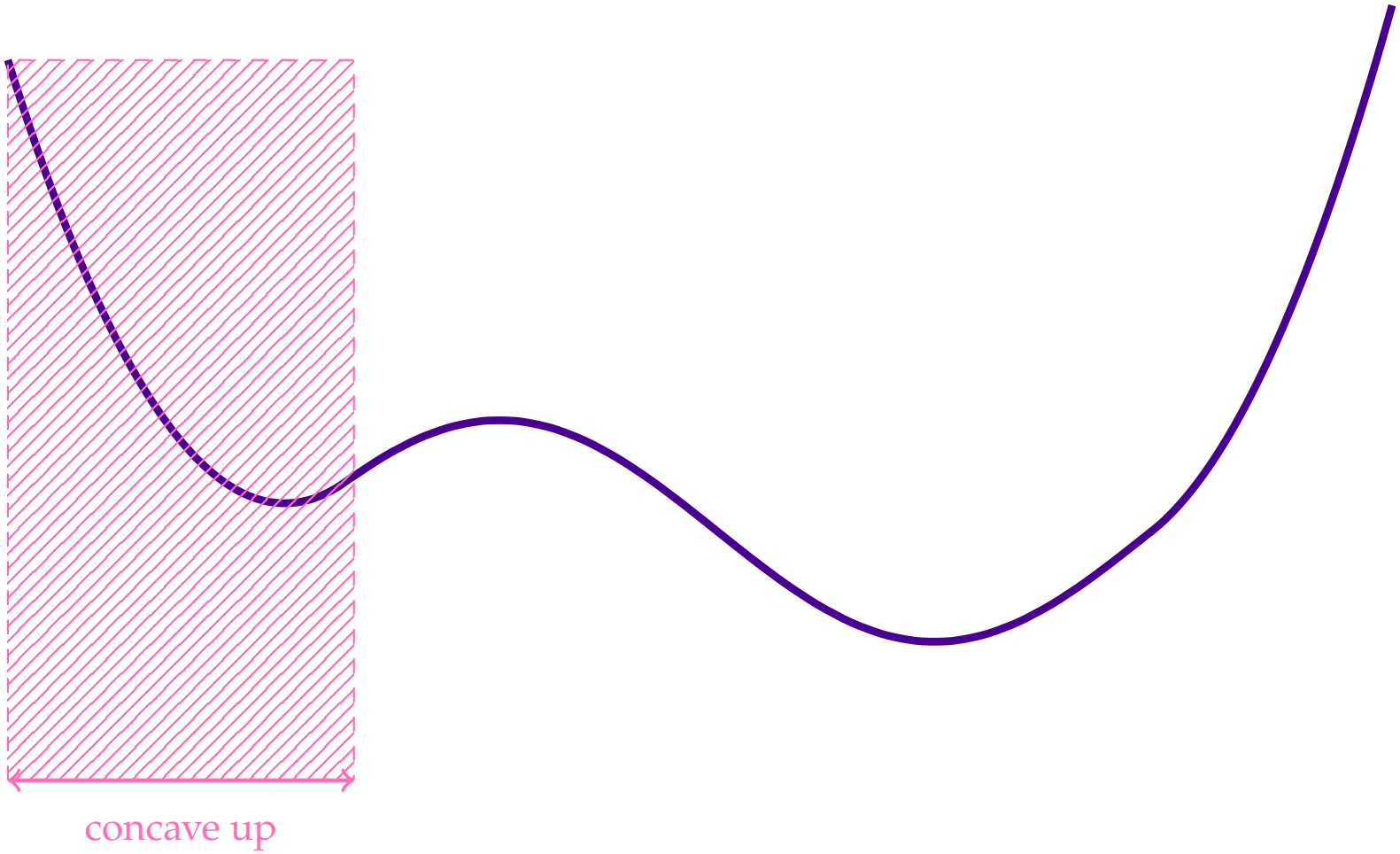
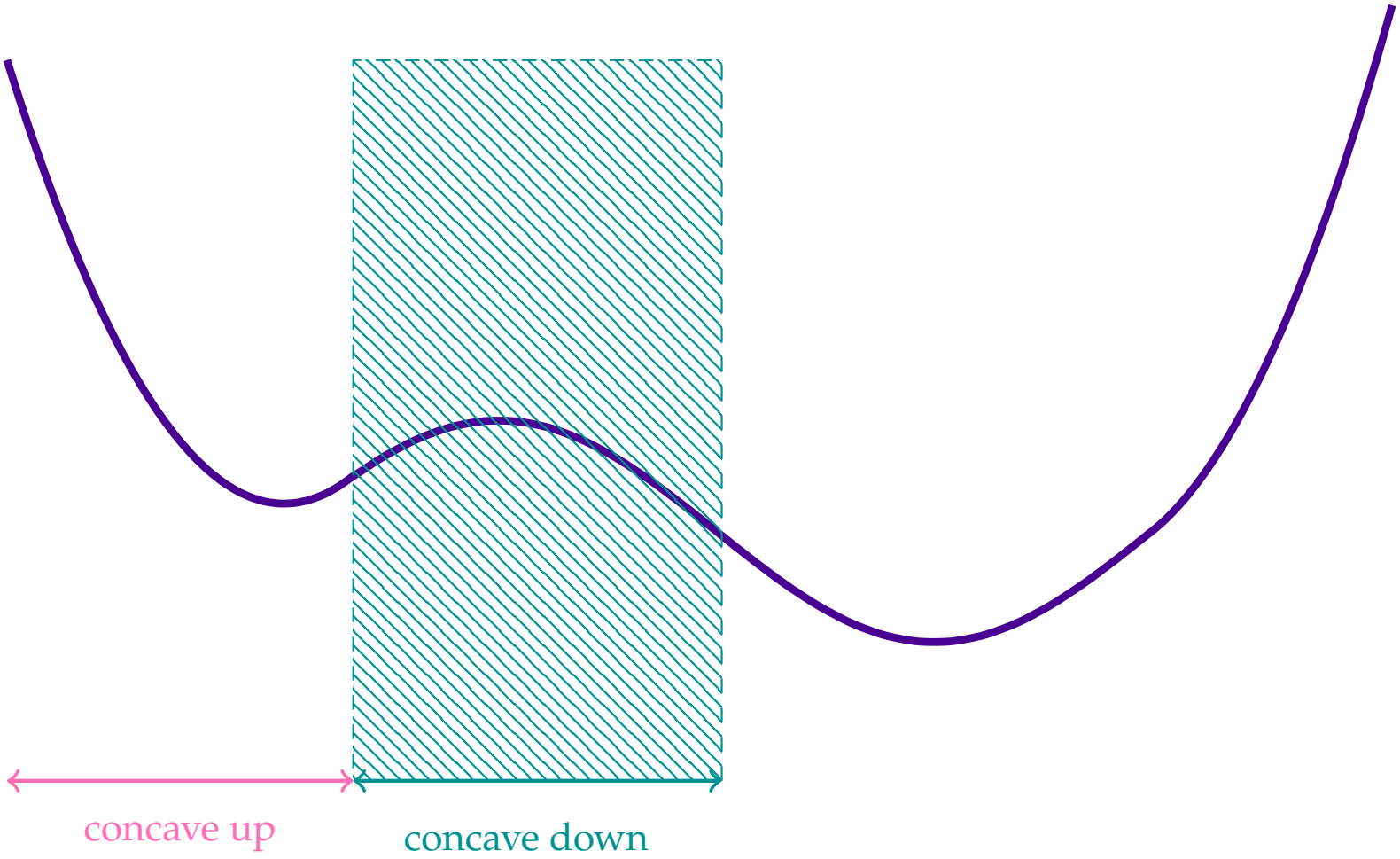
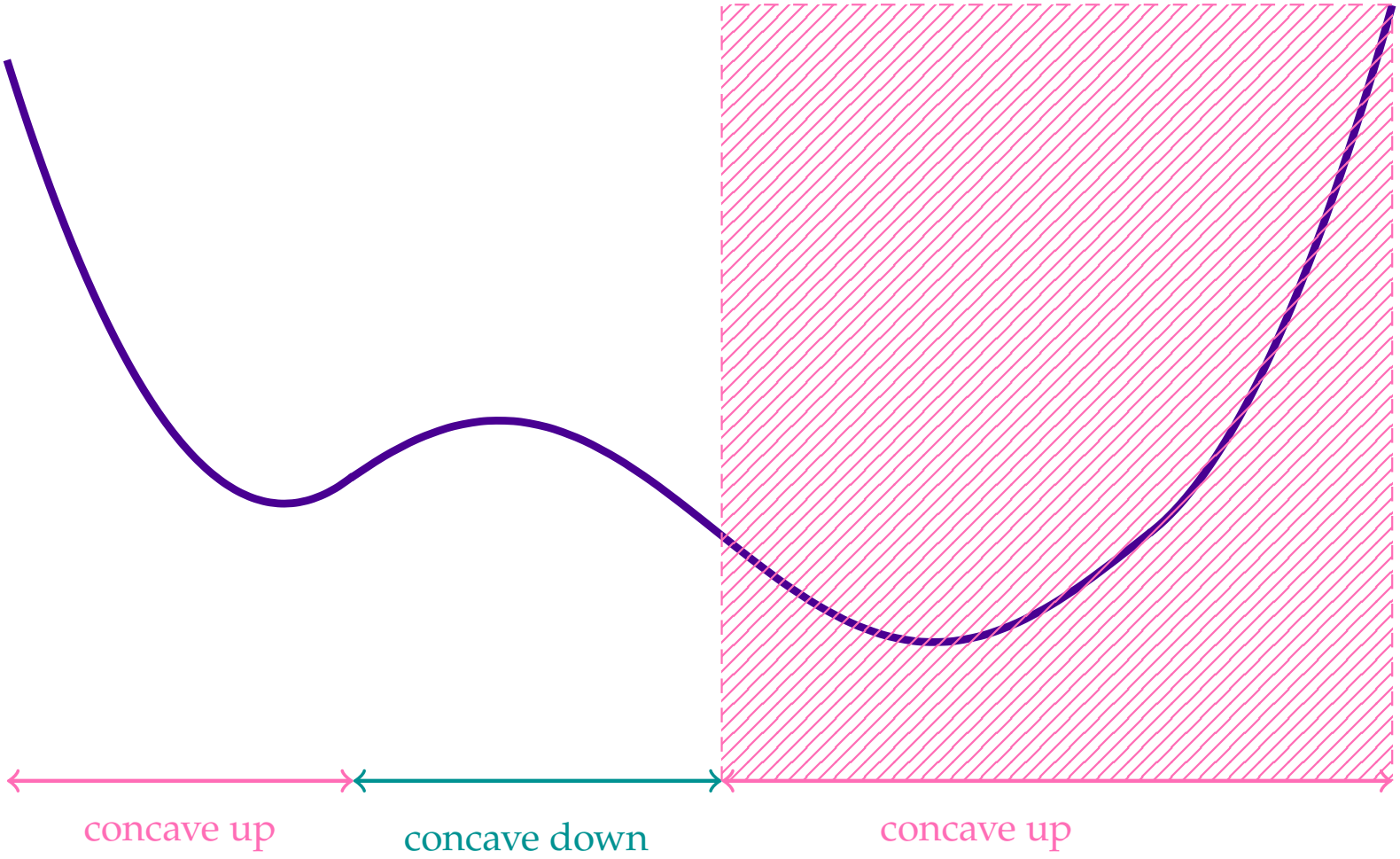
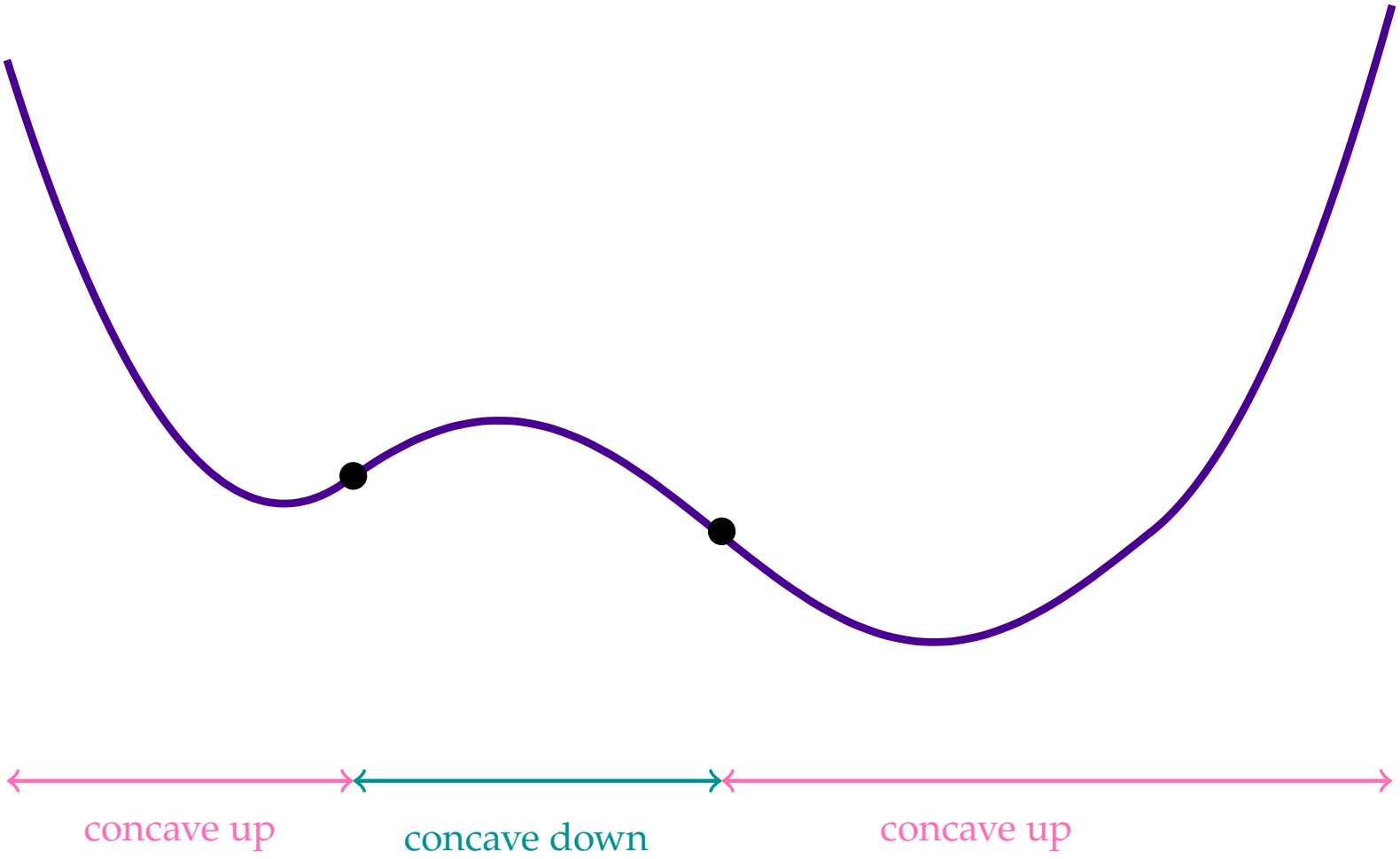
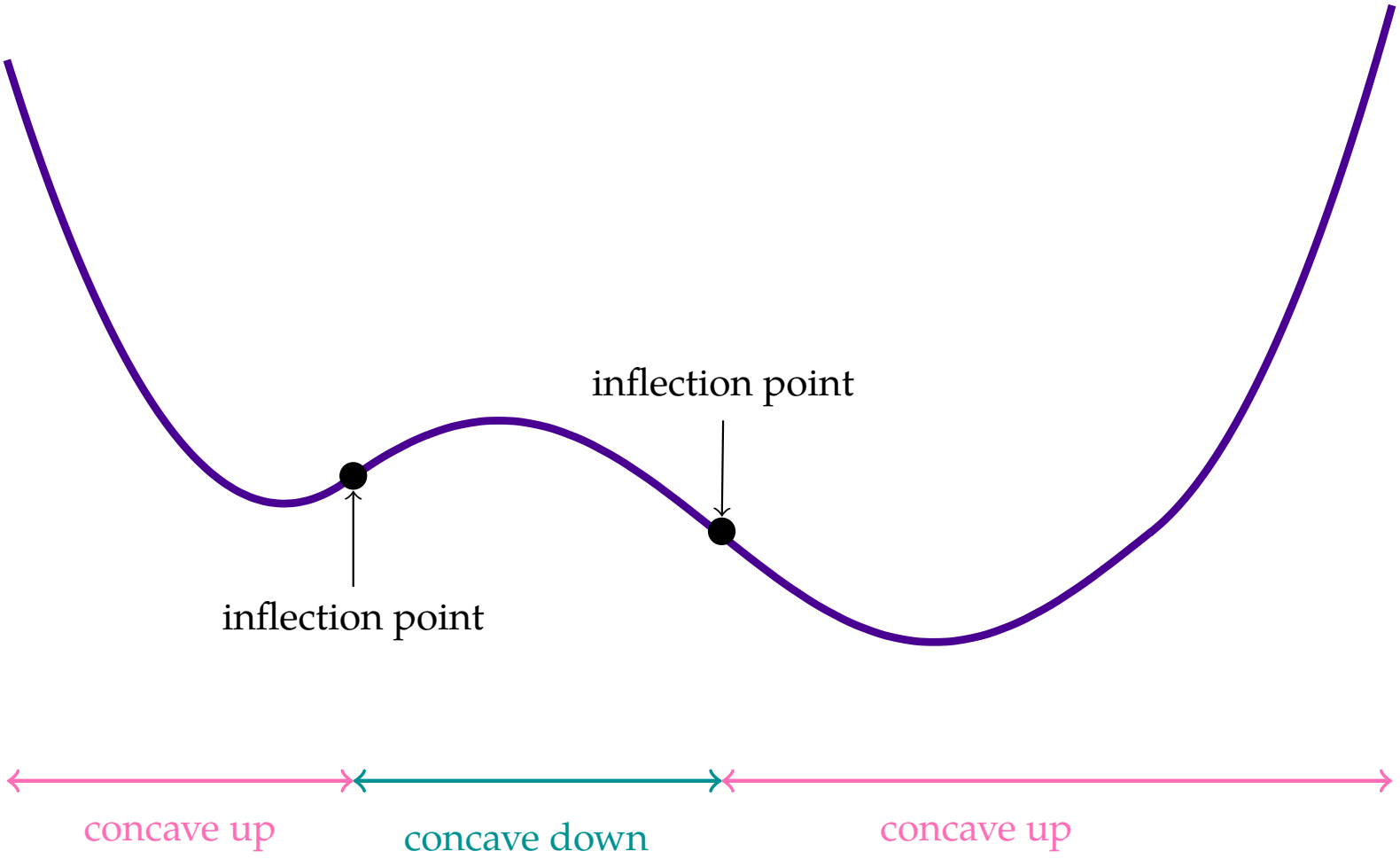


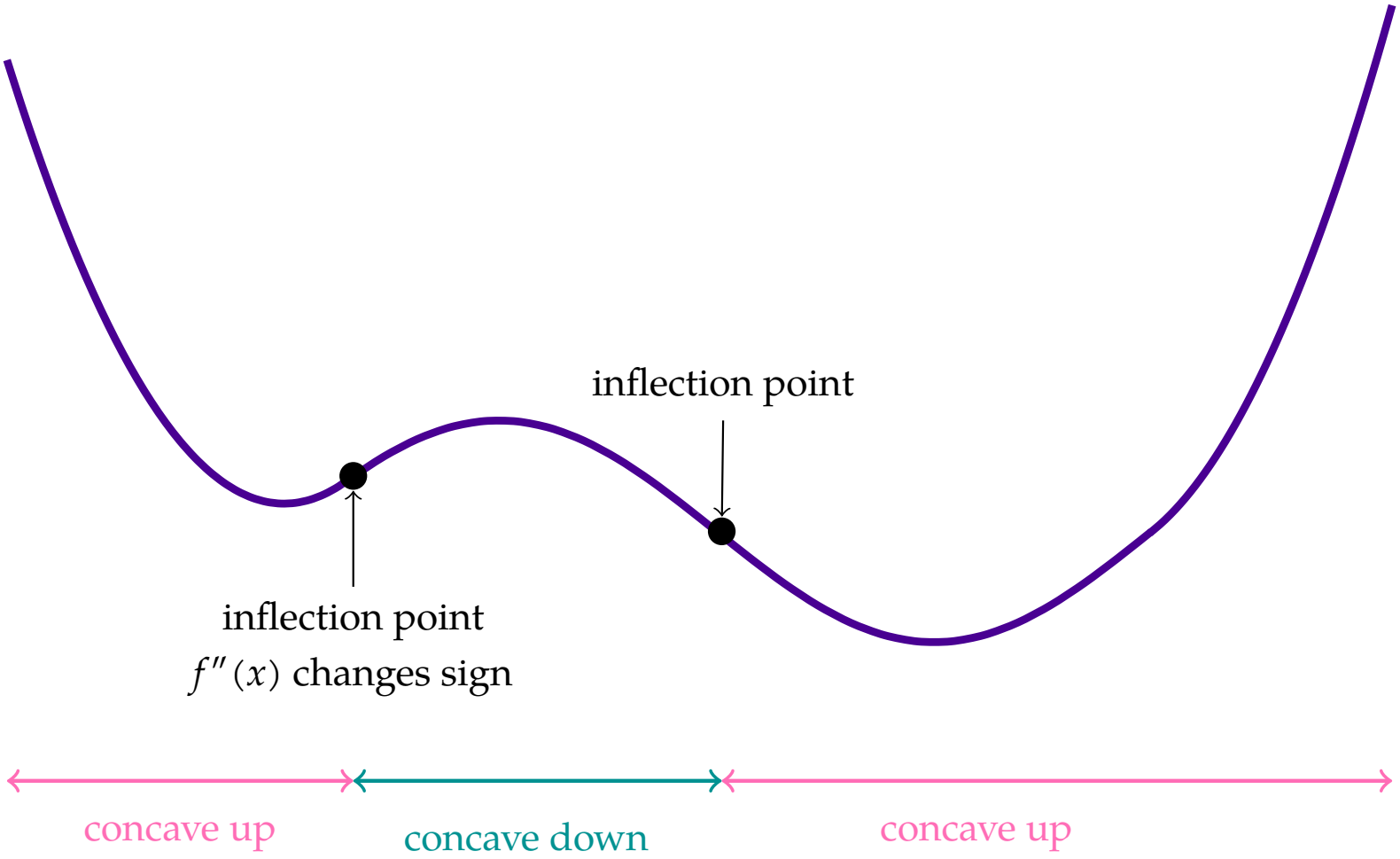

concave up


concave up
concave down


concave up
concave down
concave up


concave up
concave down
concave up


inflection point
inflection point
concave up
concave down
concave up


inflection point
inflection point
$f''(x)$ changes sign
concave up
concave down
concave up

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b),

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b),

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a, b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h)$, $h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\ (a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) = 0$,

The First Derivative Test I: Let f be continuous at x_0 and x_0 is not an endpoint of f, then

1. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) > 0$ on (a, x_0) and $f'(x) < 0$ on (x_0, b), then f has a local maximum value at x_0.
2. If $\exists\,(a,b)$ containing x_0 s.t. $f'(x) < 0$ on (a, x_0) and $f'(x) > 0$ on (x_0, b), then f has a local minimum value at x_0.

The First Derivative Test II: Let a be the left endpoint of the domain of f and f is right continuous at a.

1. If $f'(x) > 0$ on $(a, a+h), h > 0$, then f has a local minimum value at a.
2. If $f'(x) < 0$ on $(a, a+h), h > 0$, then f has a local maximum value at a.

The First Derivative Test III: Let b be the right endpoint of the domain of f and f is left continuous at b.

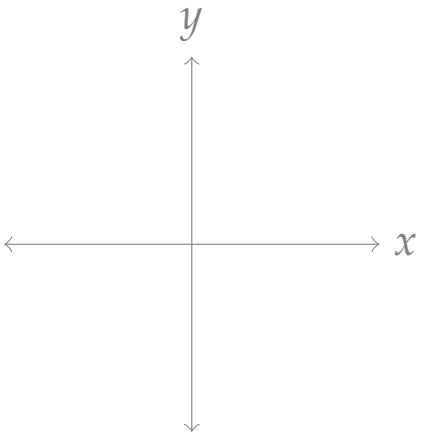
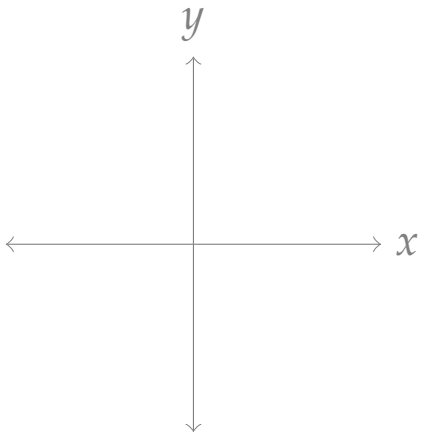
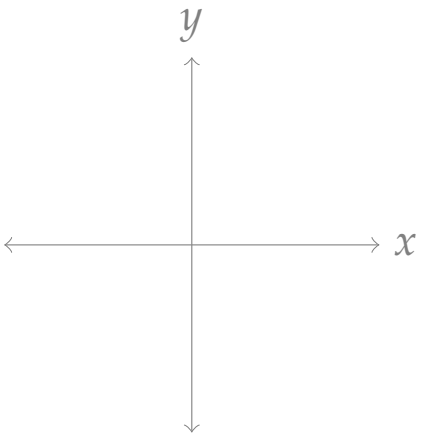
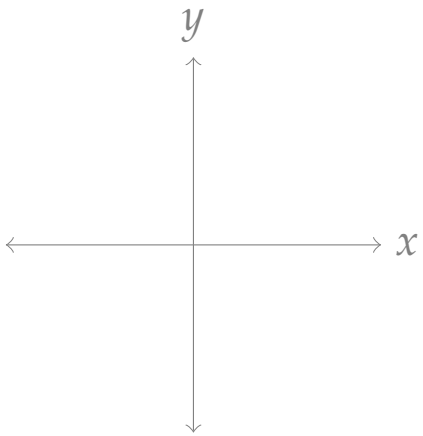
1. If $f'(x) > 0$ on $(b-h, b), h > 0$, then f has a local maximum value at b.
2. If $f'(x) < 0$ on $(b-h, b), h > 0$, then f has a local minimum value at b.

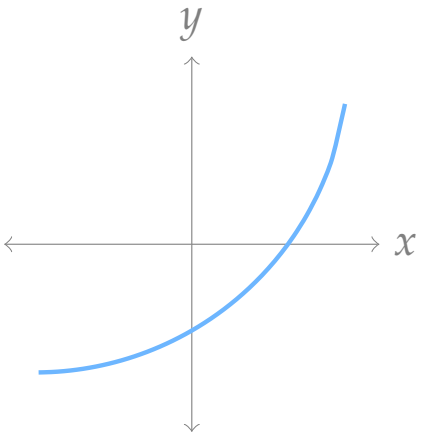
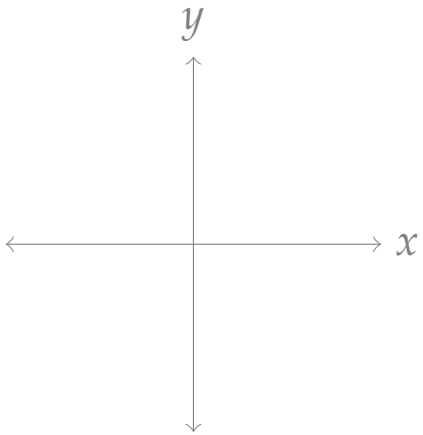
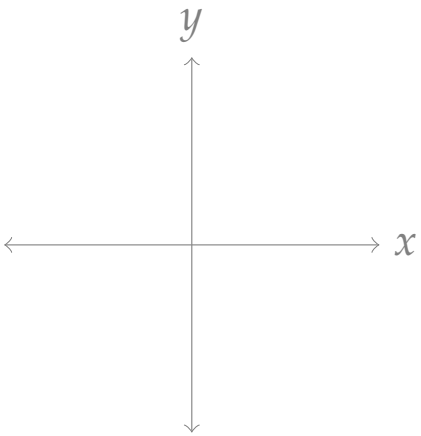
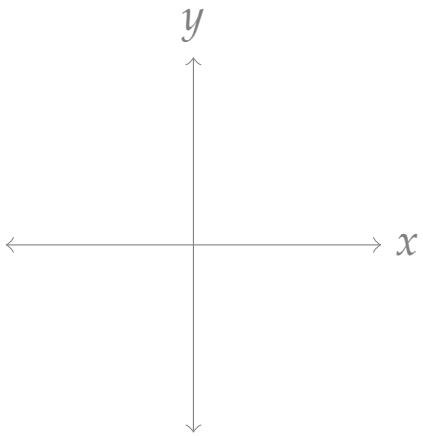
The Second Derivative Test

If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local maximum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local minimum value at x_0.

If $f'(x_0) = 0$ and $f''(x_0) = 0$, then no conclusion can be drawn.

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

	concave up ($f'' > 0$)	concave down ($f'' < 0$)
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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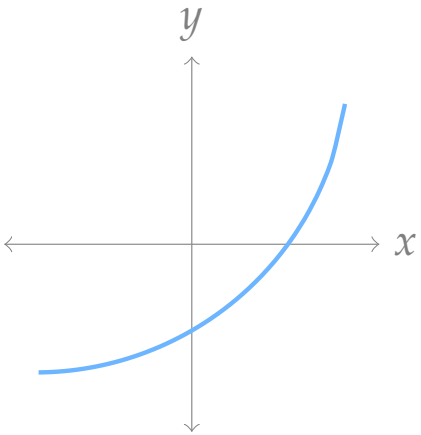
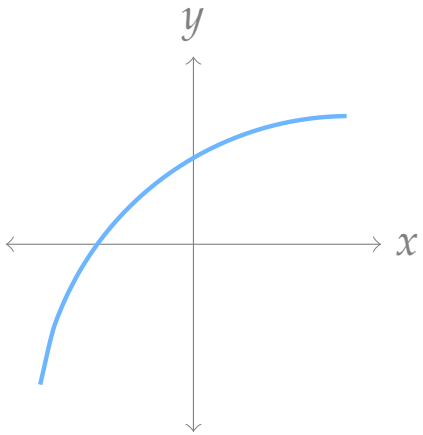
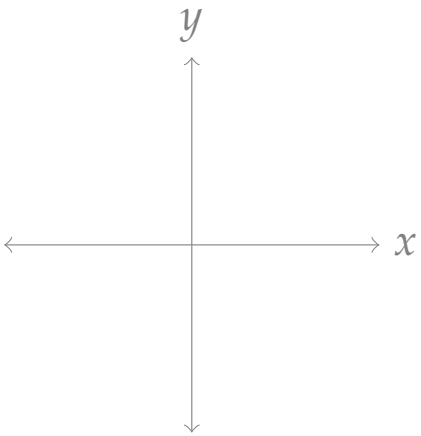
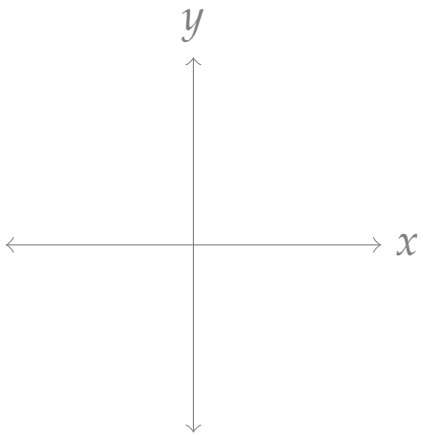
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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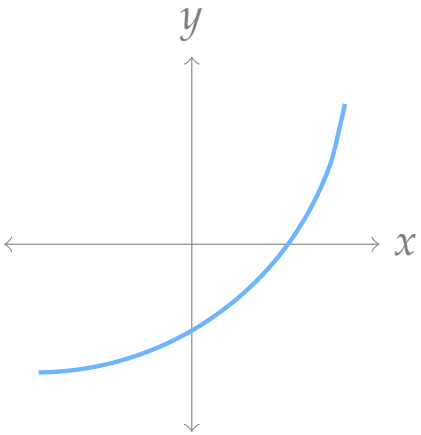
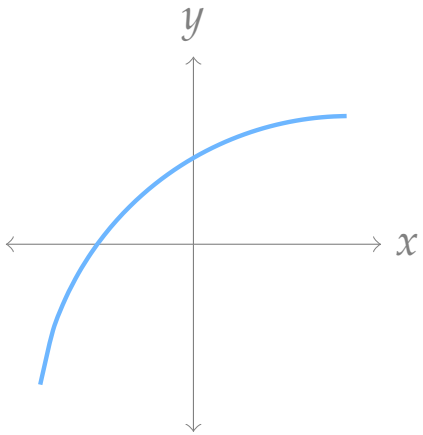
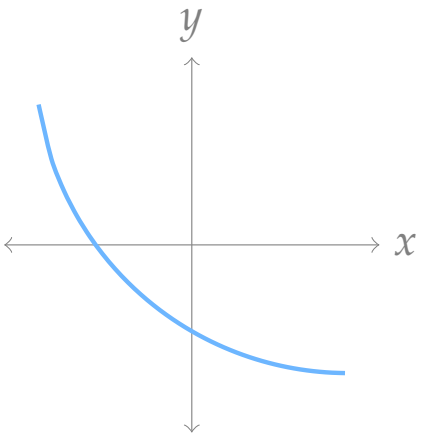
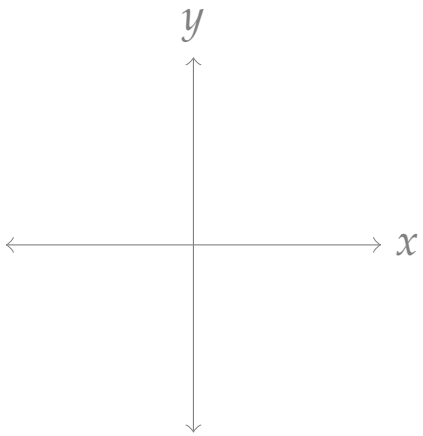
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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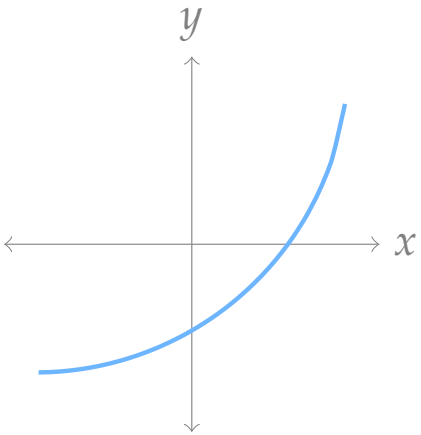
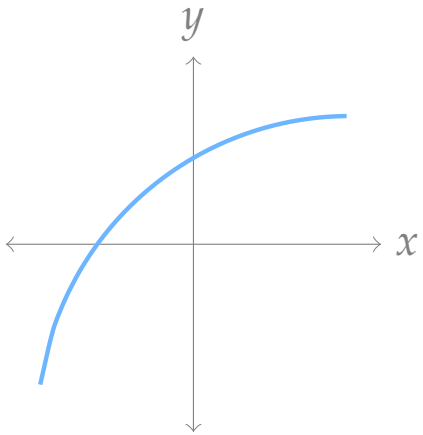
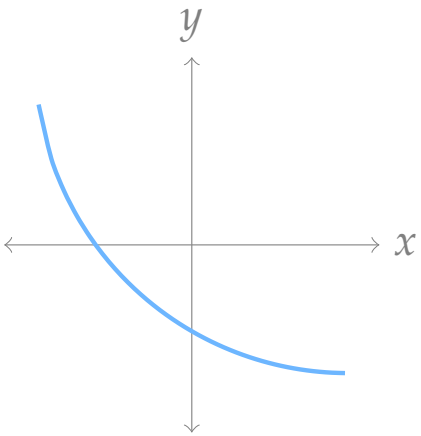
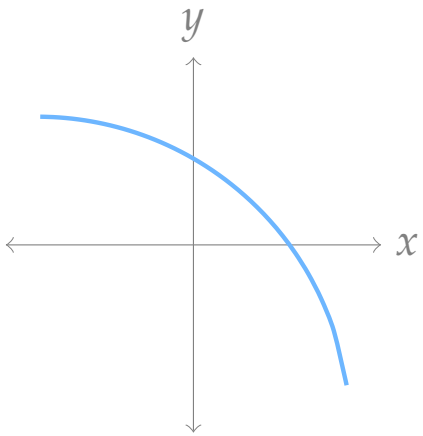
increasing ($f' > 0$)		
decreasing ($f' < 0$)		

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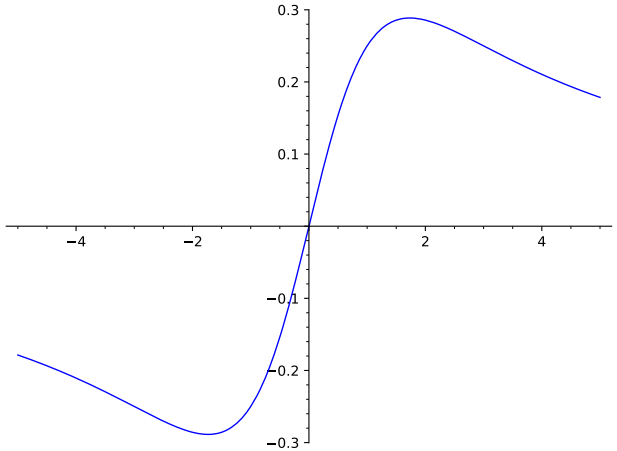
inflection points: $f'' = 0$ and of different sign before and after. $x = -3, 0, 3$.

Sketch the graph:



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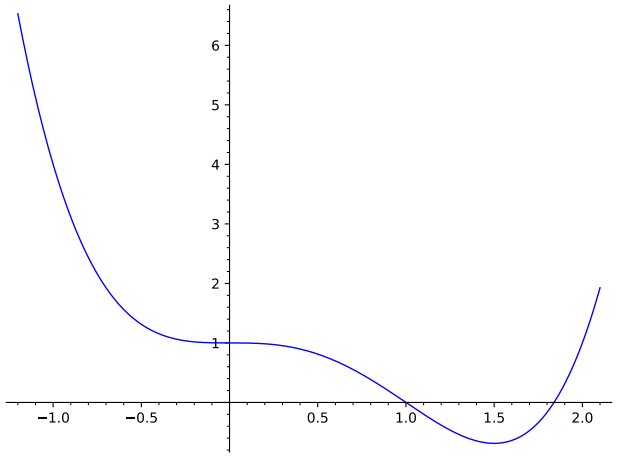

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	$+$		$-$		$-$		$-$		$-$		$+$	f'
$-$		$-\sqrt{3}$	$-$	-1	$+$	0	$-$	1	$+$	$\sqrt{3}$	$+$	f''

singular points: $x^2 - 1 = 0$. vertical asymptotes $x = \pm 1$.

asympote: $y = x$, for $\lim_{x\to\pm\infty}\left(\frac{x^3}{x^2-1} - x\right) = 0$

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interval of decrease:

Example III

Determine the intervals of increase and decrease, the local extreme values, the concavity and the graph of $f(x) = \frac{x^3}{x^2-1}$.

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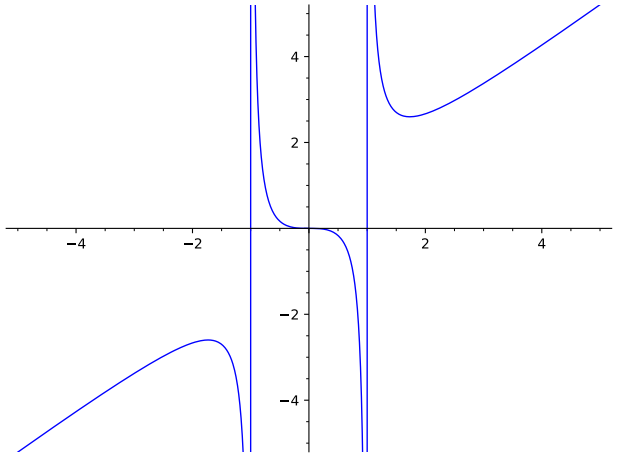
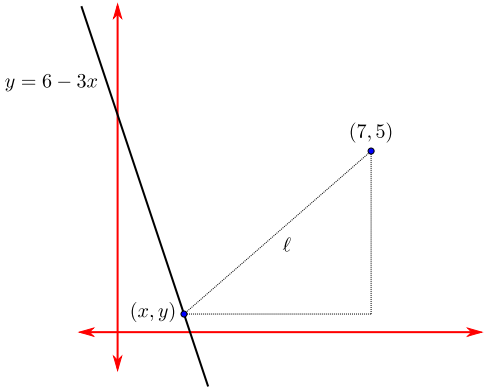


Extreme-Value Problems

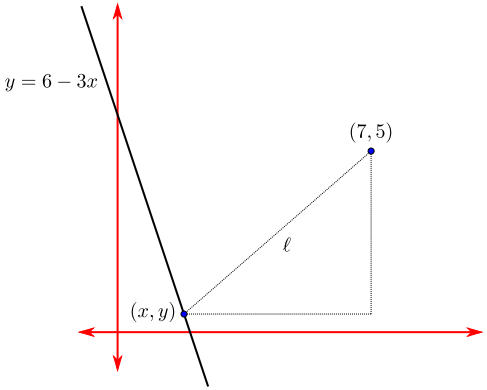
Example I



Find the point on the line $y = 6 - 3x$ that is closest to $(7, 5)$ and the shortest distance.

Review of the Key Theorem

Example I



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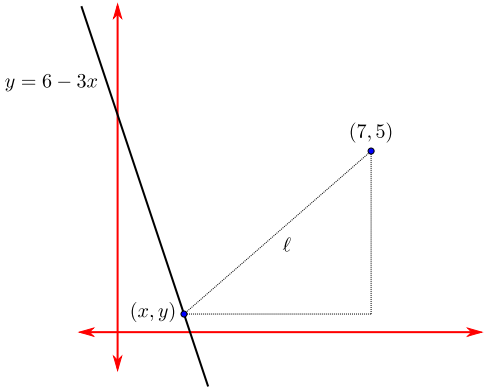

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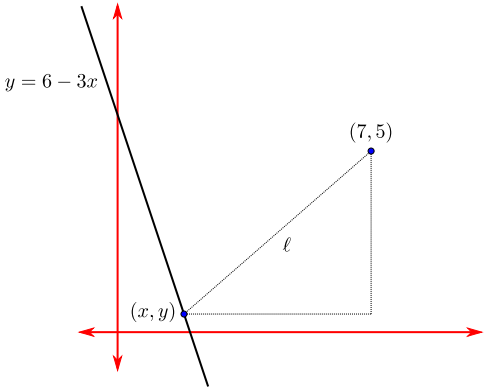

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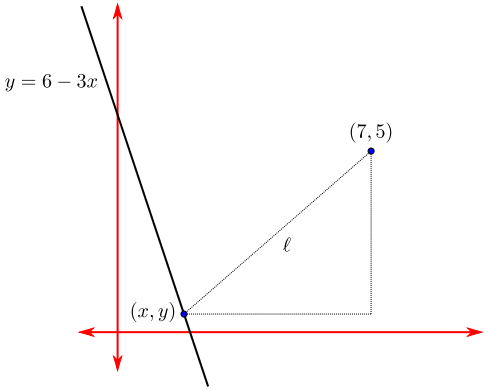

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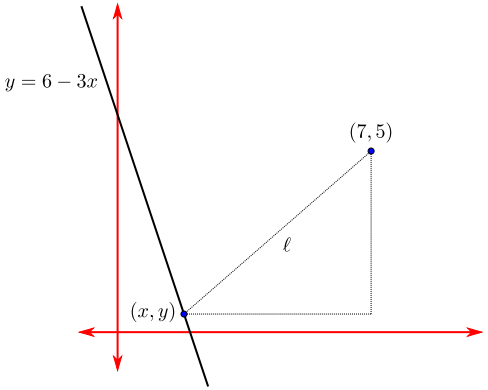

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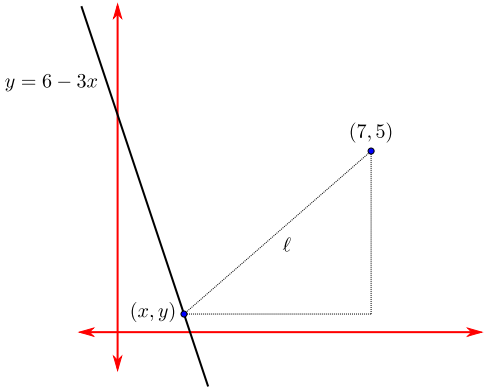

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$$(\ell^2(x))' = 10(2x - 2)$$

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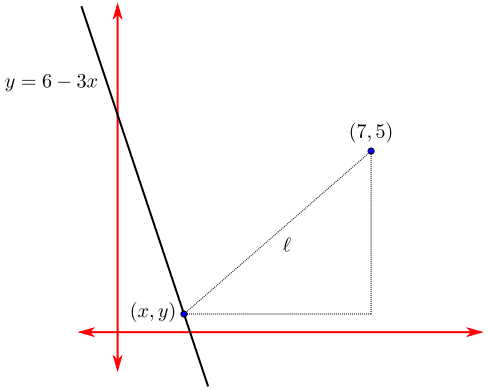

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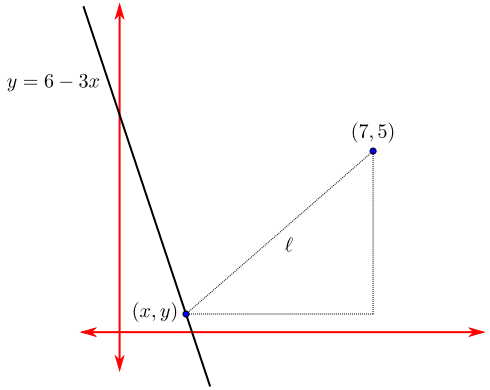

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The minimal distance is $\ell(1) = 2\sqrt{10}$ and occurs at $x = 1, y = 6 - 3 = 3$.

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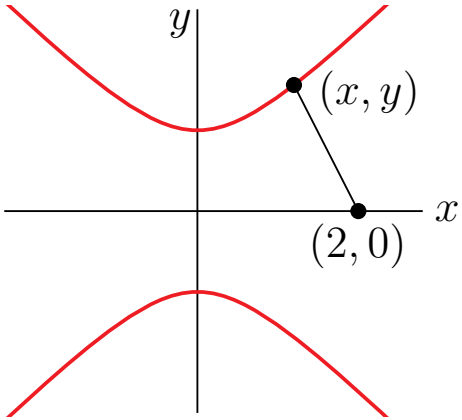

Find the point on $y^2 = x^2 + 1$ that is closest to $(2, 0)$ and the shortest distance.

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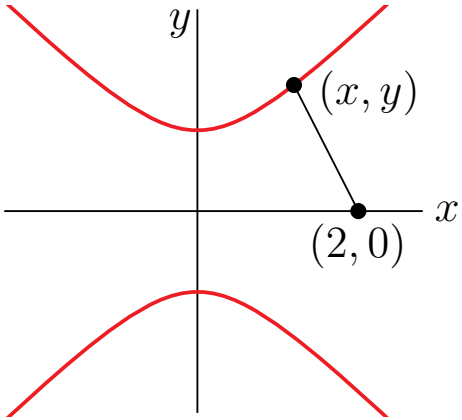

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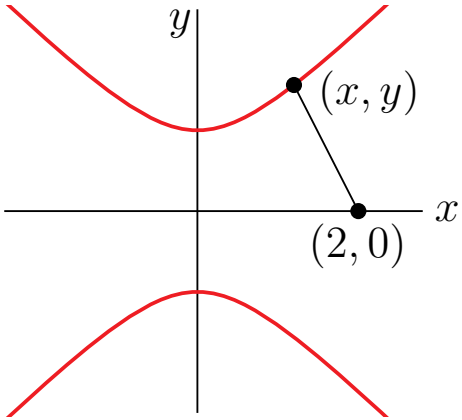

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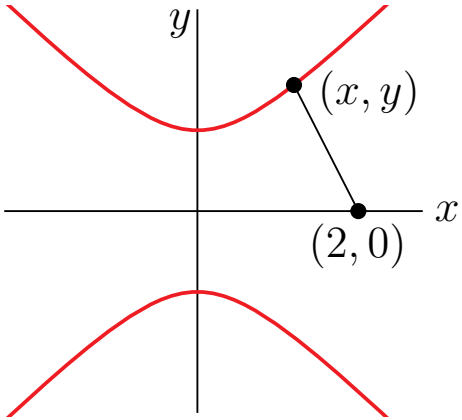

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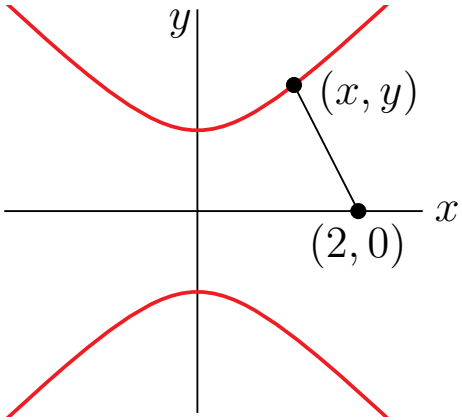

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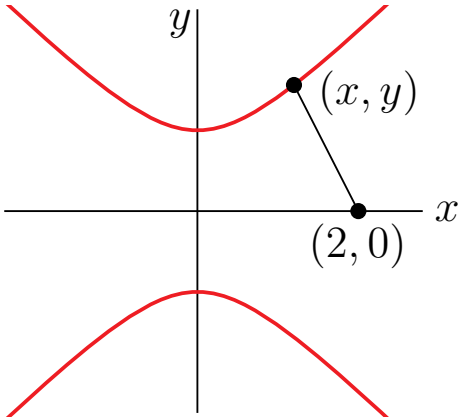

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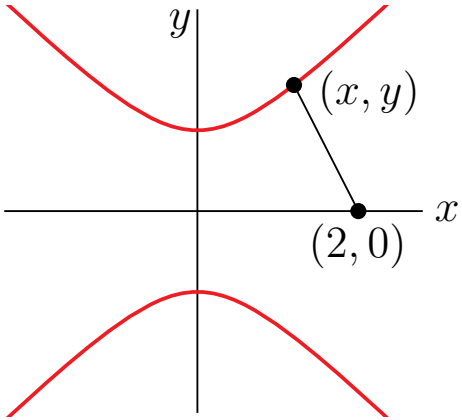

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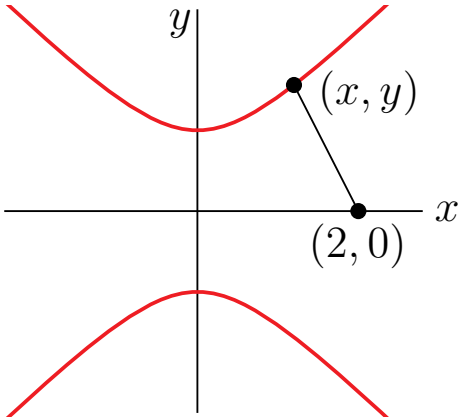

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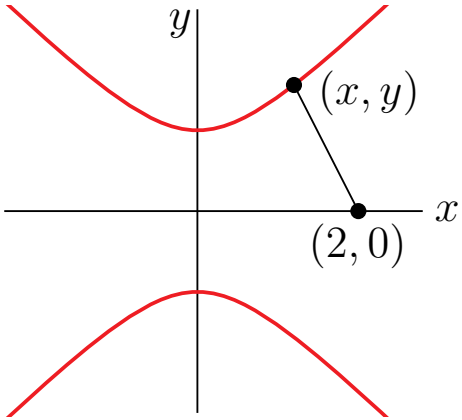

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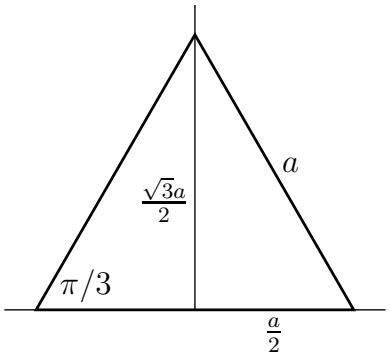
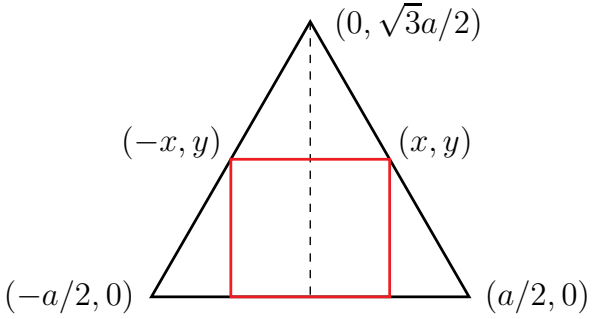
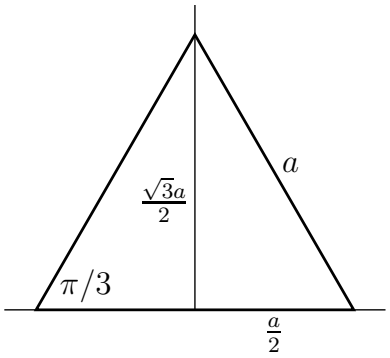
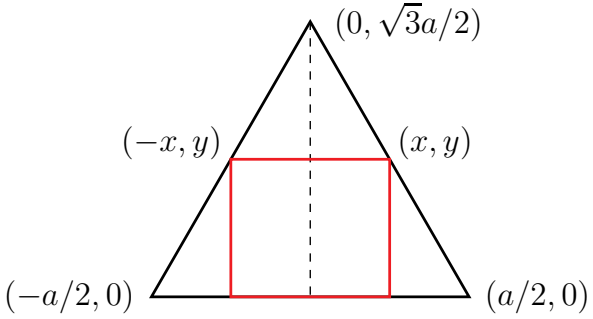




Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side a if one side of the rectangle lies on the base of the triangle.

Example III





Find the dimensions of the rectangle of largest area that can be inscribed in an equilateral triangle of side a if one side of the rectangle lies on the base of the triangle.

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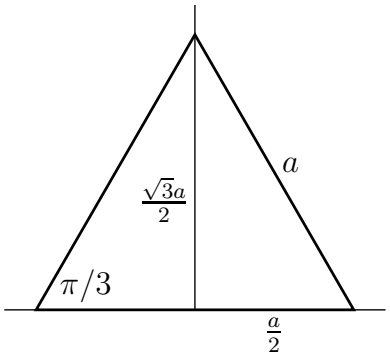

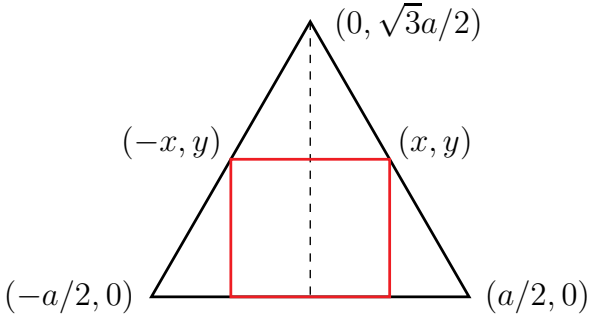

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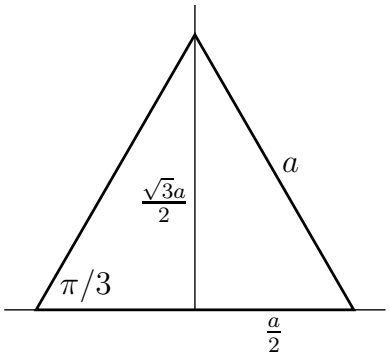

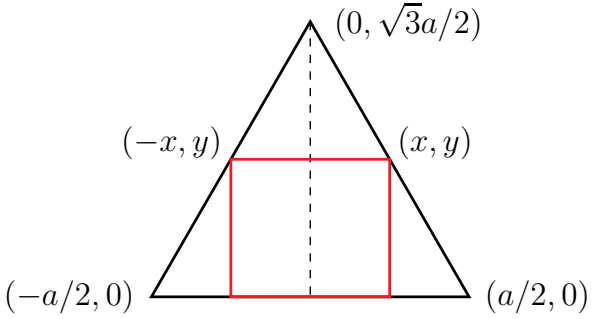

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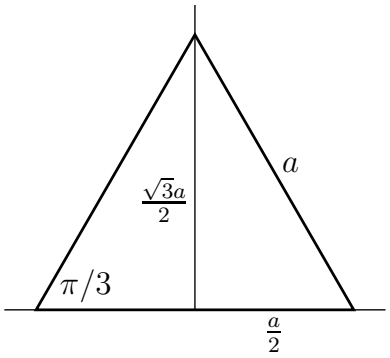

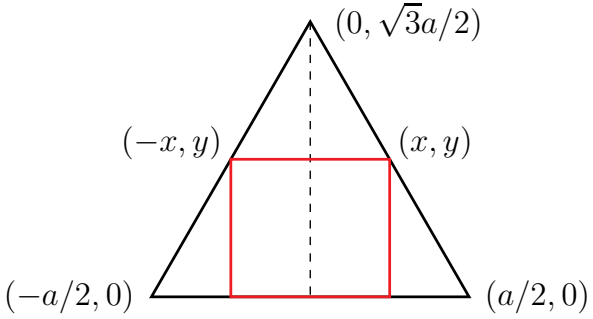

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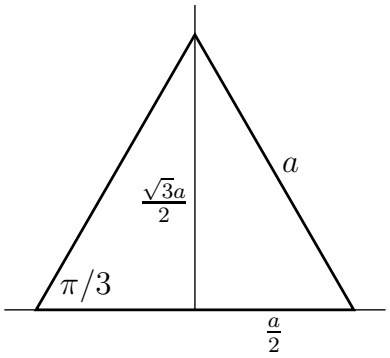

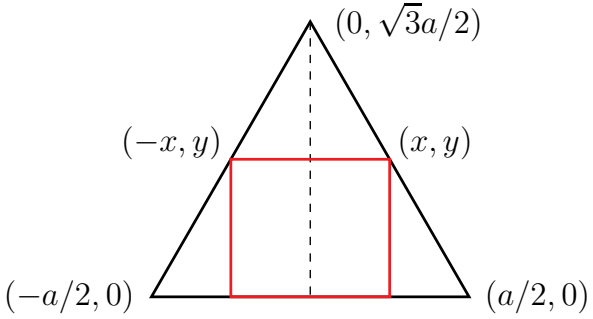

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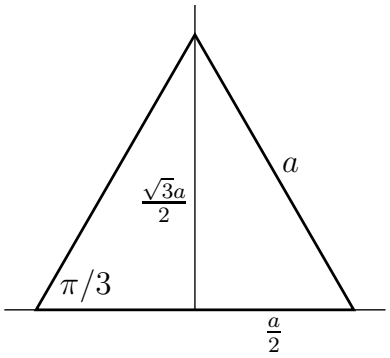

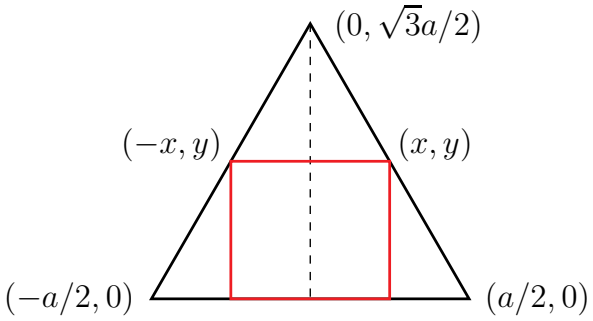

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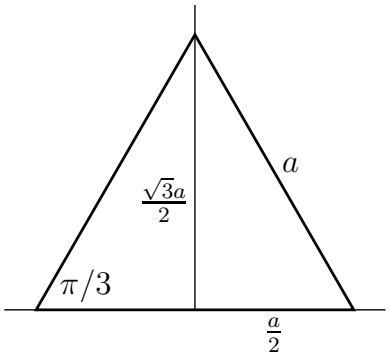

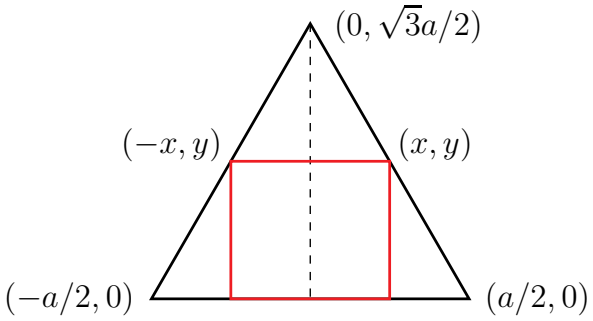

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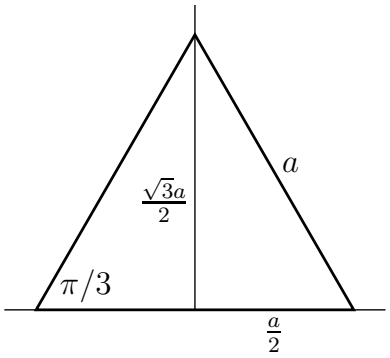

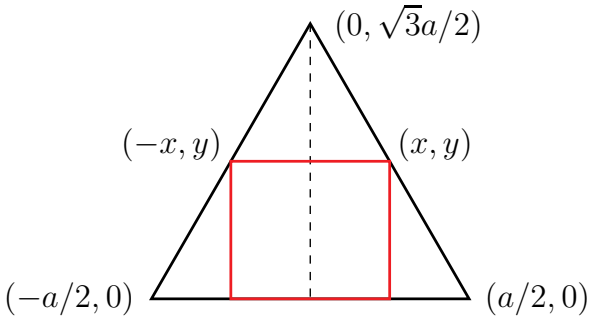

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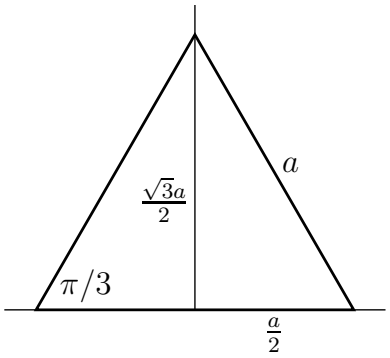
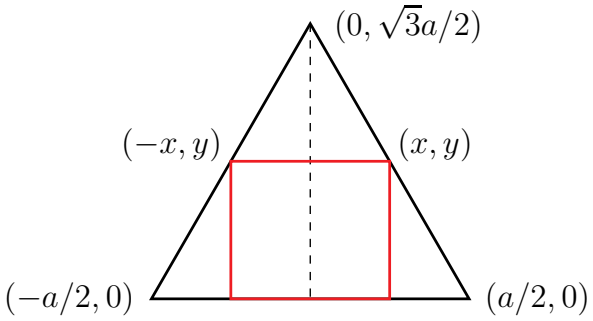
As $x = 0, \frac{a}{2}$, $A(x) = 0$, and $A(\frac{a}{10}) > 0$, so the absolute maximum exists.





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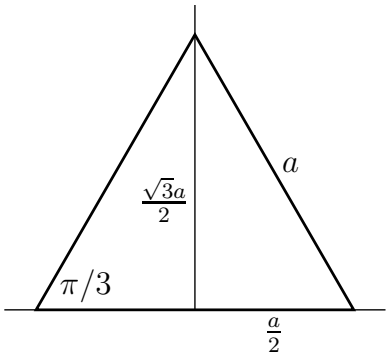
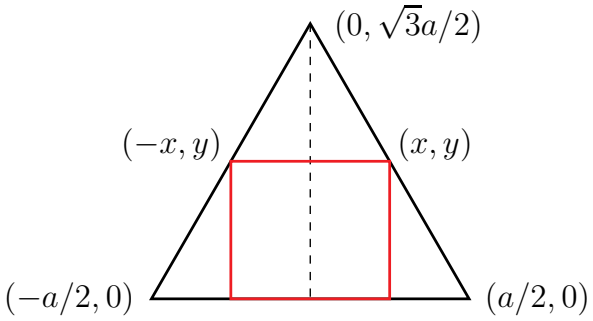
Compute the derivative of $A(x)$:





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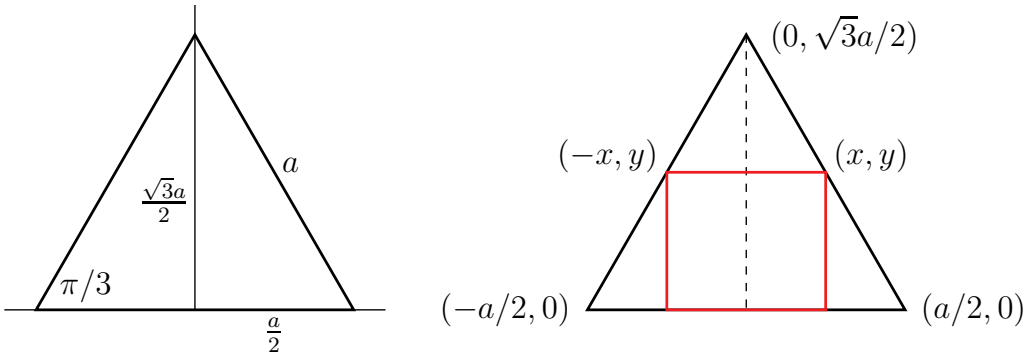

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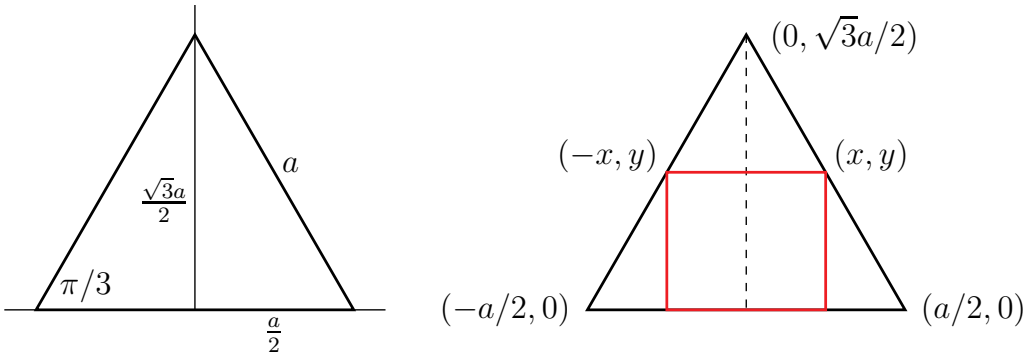
The minimizer $x^\star$ is the root of $A'(x) = 0$, i.e. $x^\star = \frac{a}{4}$.

Example III



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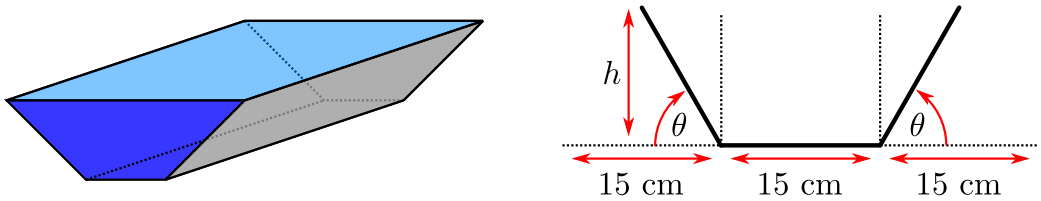
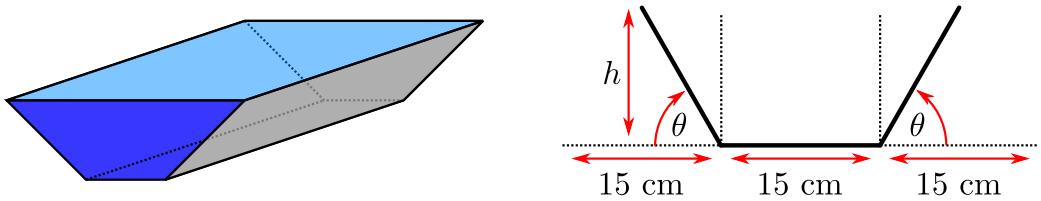
The largest such rectangle has dimensions $2 \times \frac{a}{4} = \frac{a}{2}$ by $-\sqrt{3}\,\frac{a}{4} + \frac{\sqrt{3}}{2}a = \frac{\sqrt{3}}{4}a$.

Example IV



A water trough is to be constructed from a metal sheet of width 45 cm by bending up one third of the sheet on each side through an angle θ. Which θ will allow the trough to carry the maximum amount of water?



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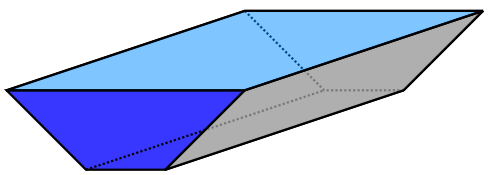
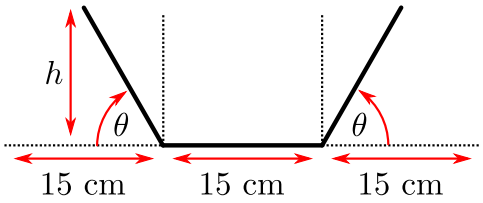
The problem amounts to maximize the cross-sectional area A.





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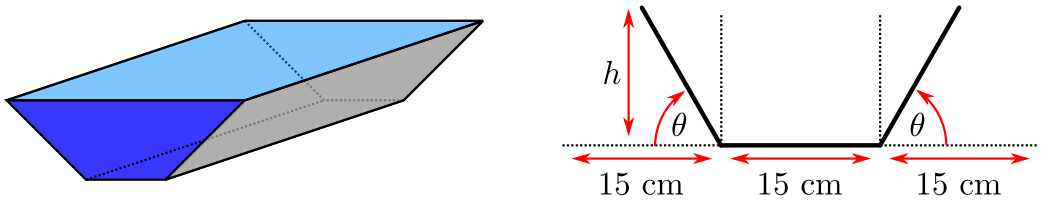
$$A(\theta) = \frac{1}{2} \times 15 \sin\theta \cdot 15 \cos\theta \times 2$$



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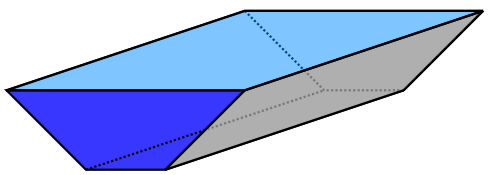

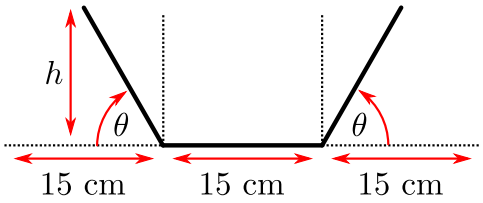

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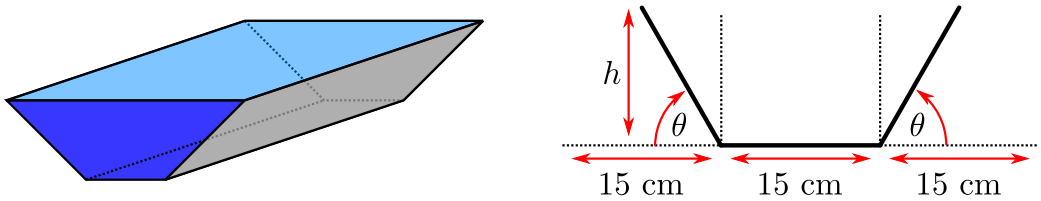
$$A(\theta) = \frac{1}{2} \times 15 \sin\theta \cdot 15 \cos\theta \times 2 + 15 \times 15 \sin\theta = 225 \sin\theta(\cos\theta + 1),$$



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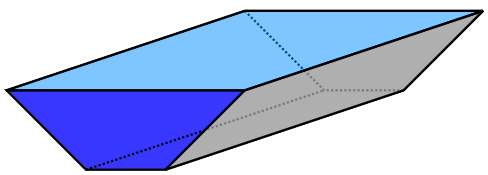

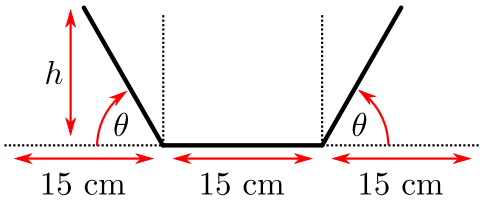

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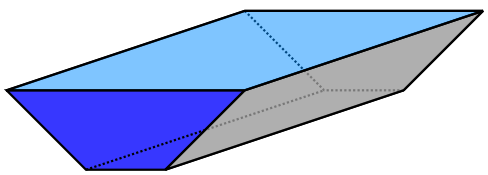
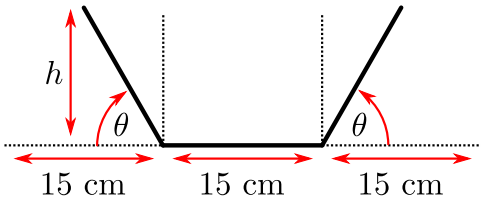
As $\theta = 0, \pi$, $A(\theta) = 0$, and $A(\frac{\pi}{2}) = 225 > 0$, so the absolute maximum exists.





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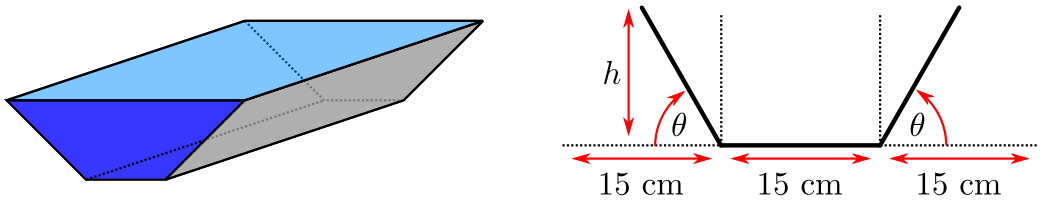

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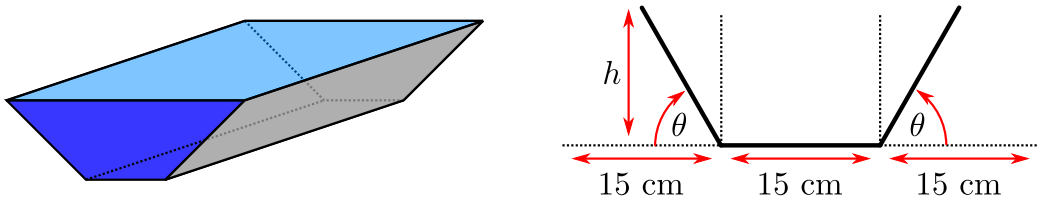

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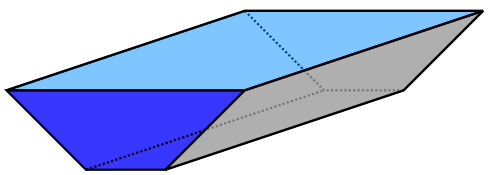

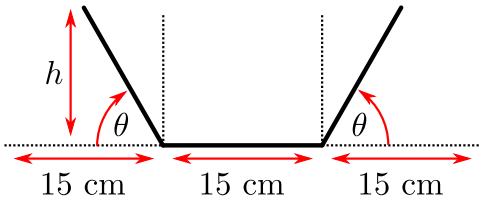

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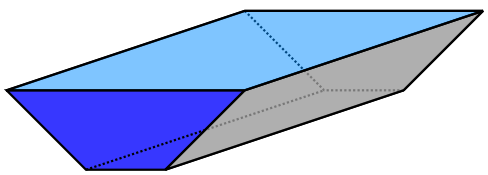

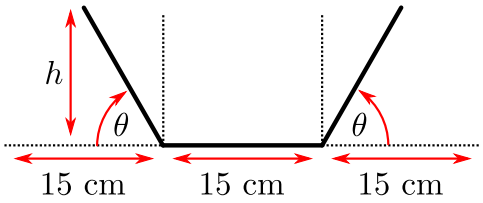

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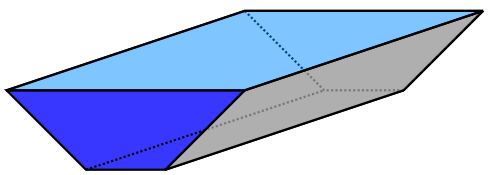

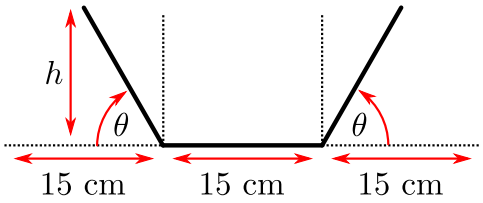

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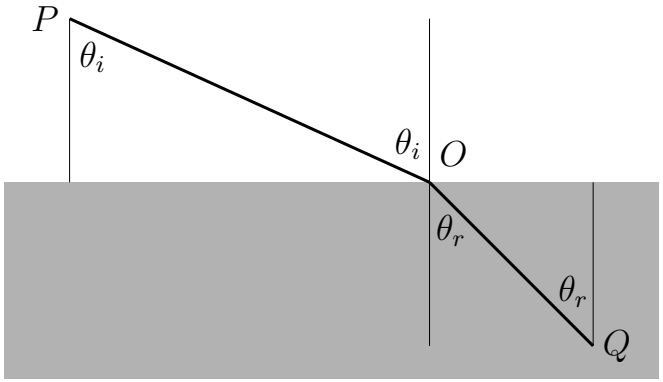
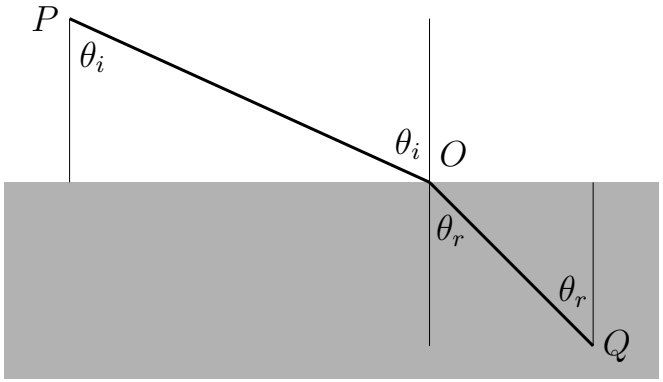
The largest area $A(\frac{\pi}{3}) = 225 \sin(\frac{\pi}{3})(\cos(\frac{\pi}{3}) + 1) = 225 \cdot \frac{3\sqrt{3}}{4} \approx 292.28$.

Example V



The Fermat principle states that the path of light travels from P to Q takes minimal time; establish the Snell's law $\frac{\sin\theta_i}{\sin\theta_r} = \frac{c_a}{c_w}$, where c_a, c_w be the speed of light in air and water.



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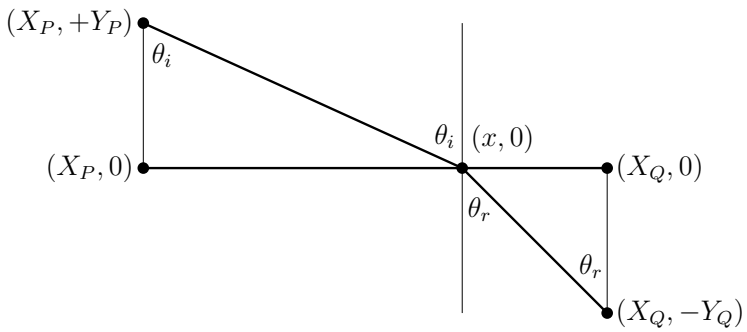

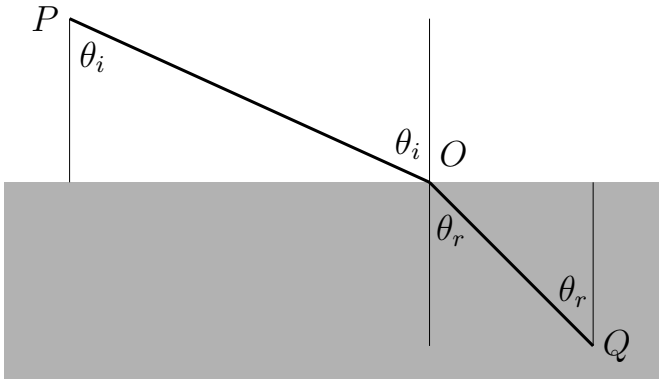

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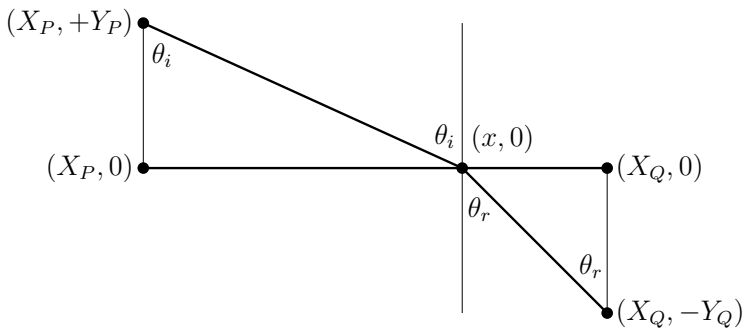

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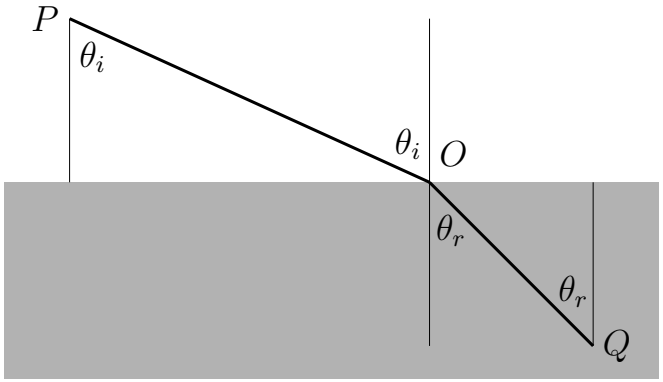

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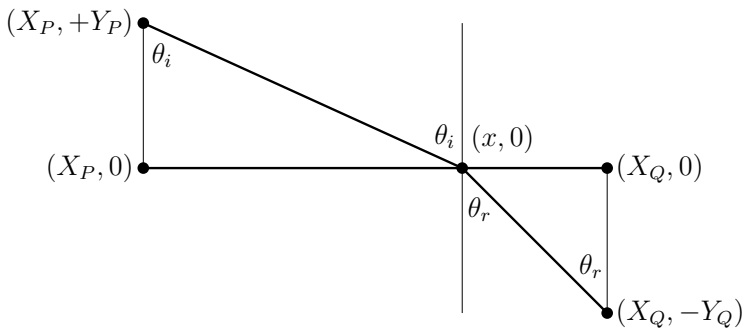

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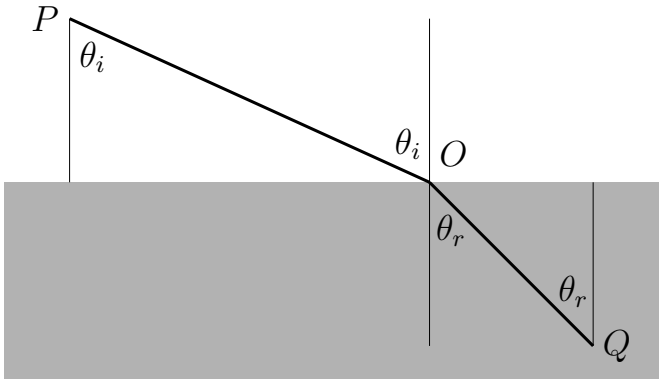

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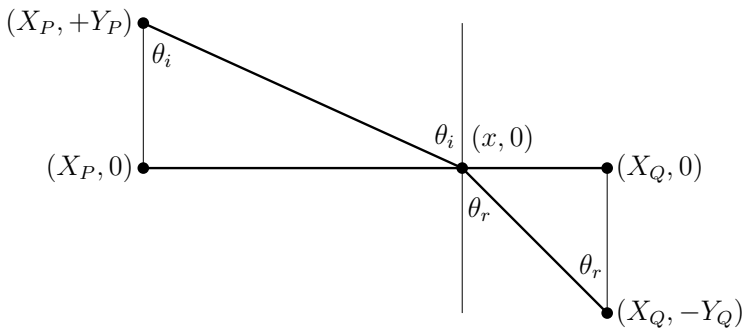

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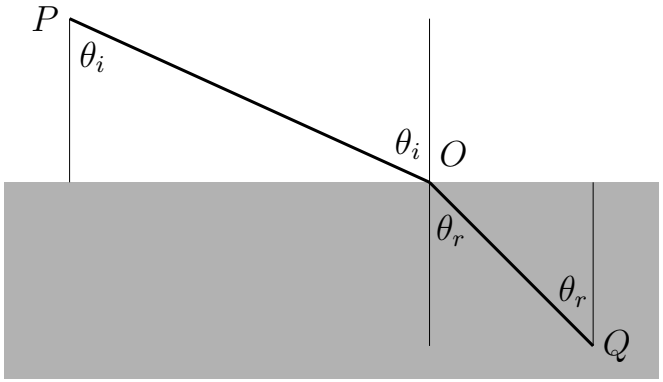

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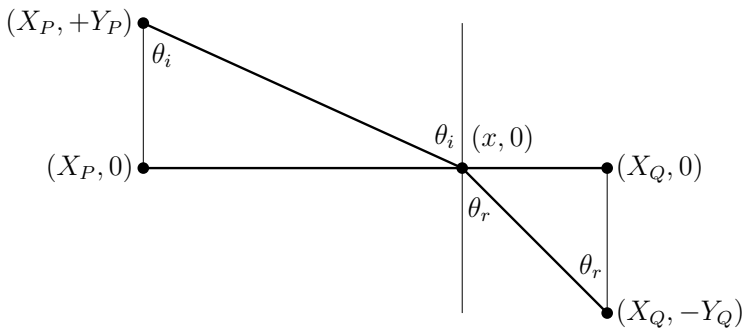

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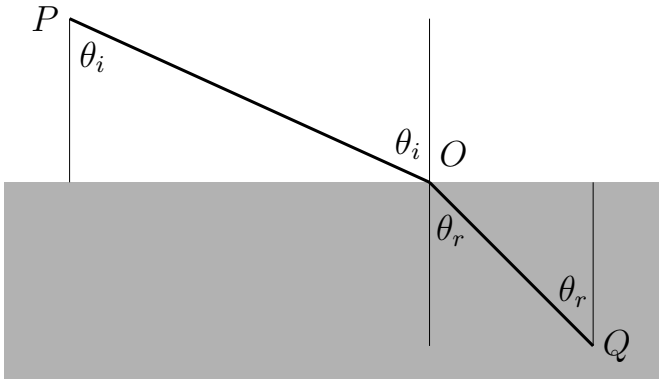

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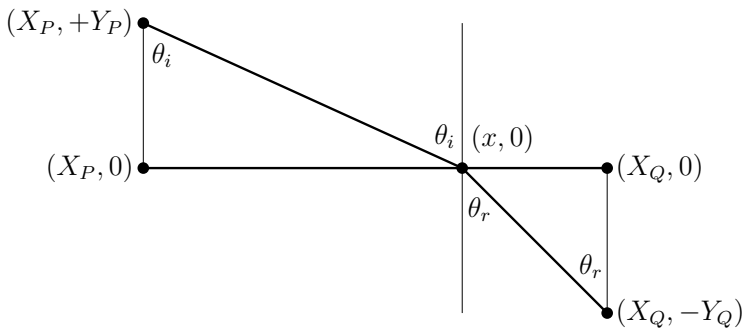

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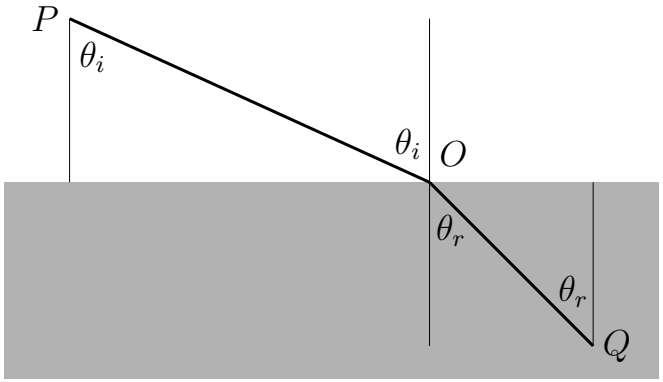

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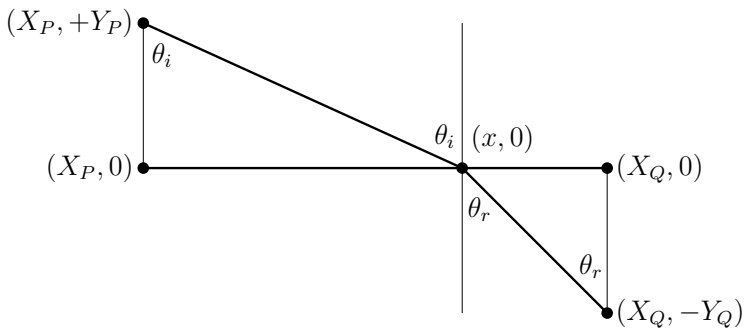

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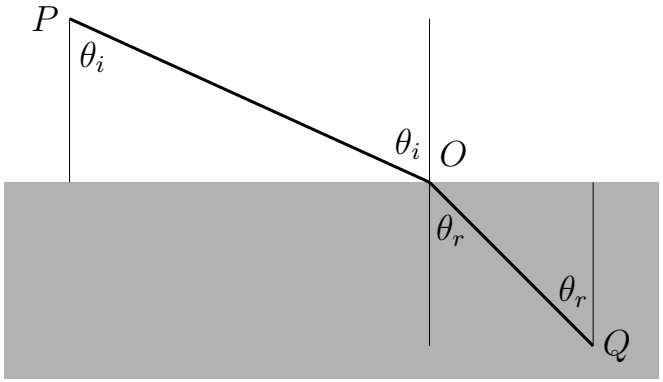

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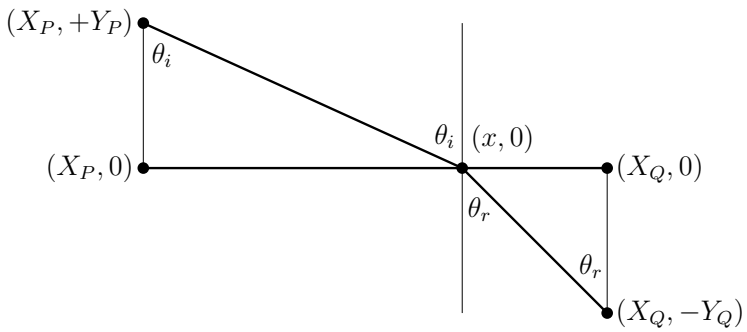

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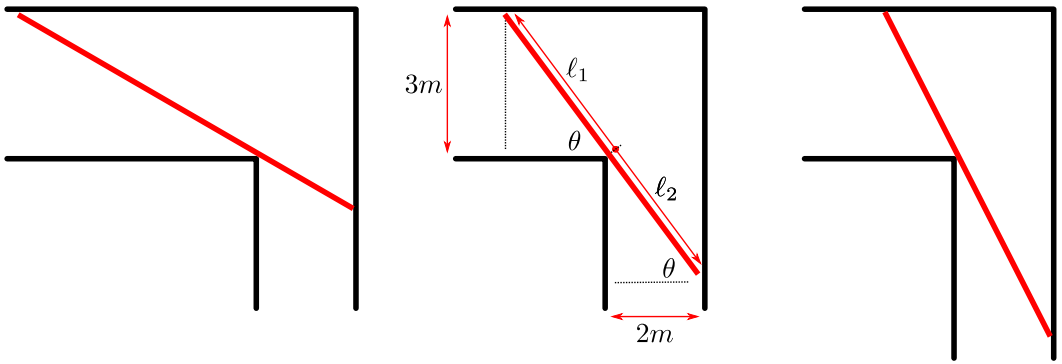
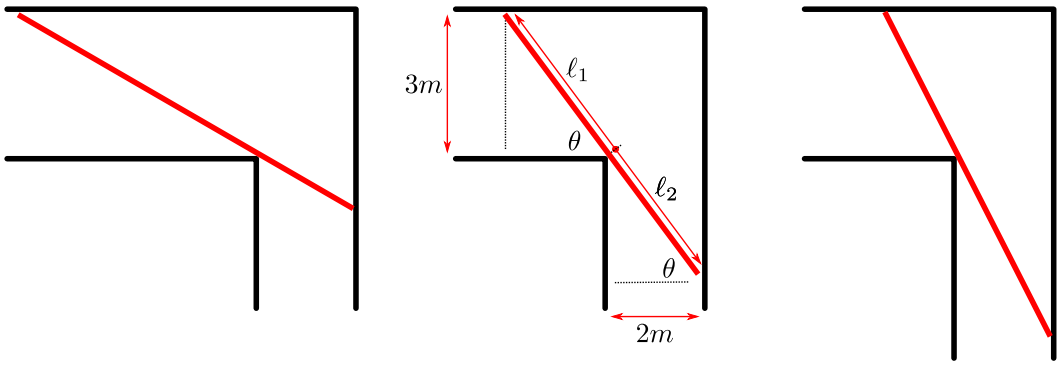
Example VI



Find the length of the longest rod that can be carried horizontally (no tilting allowed) from a corridor 3m wide into a corridor 2m wide.

Example VI



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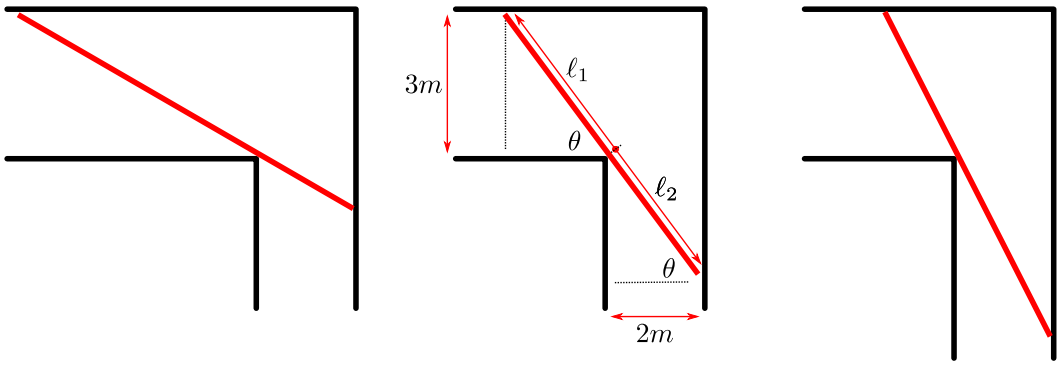

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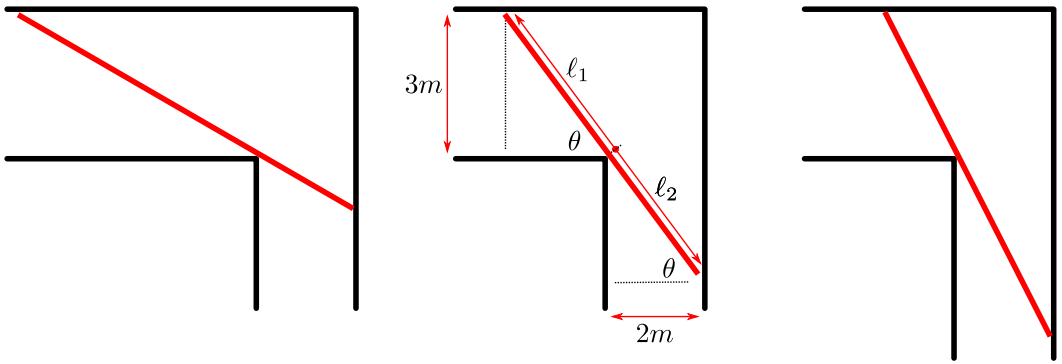

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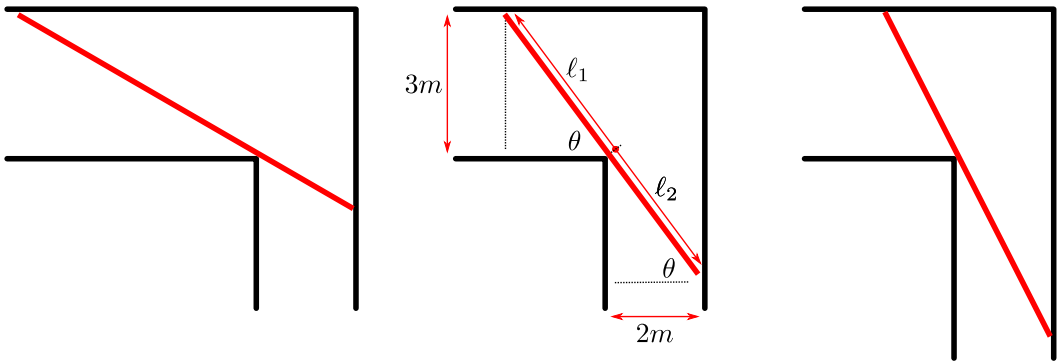

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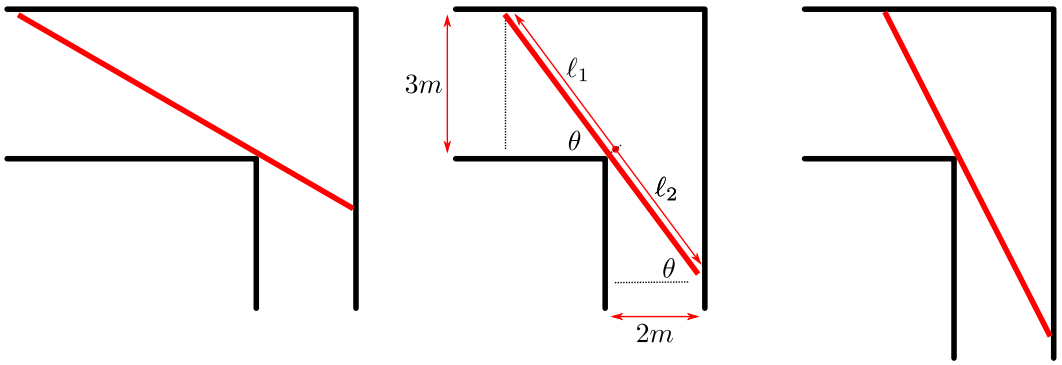

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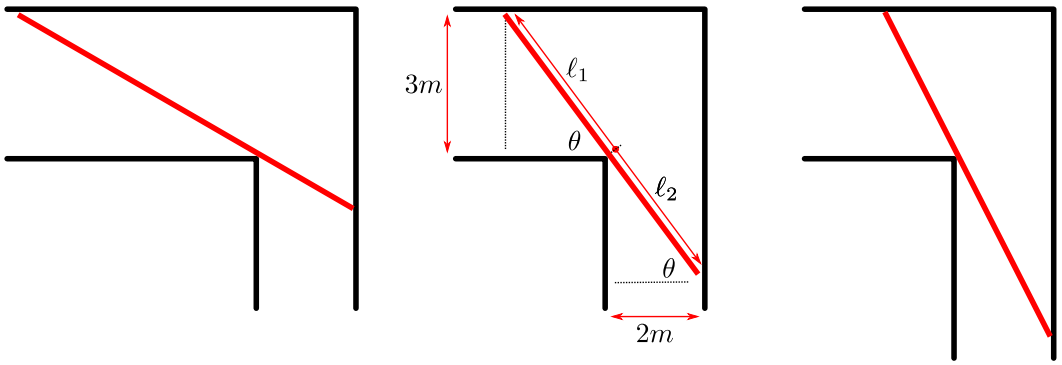

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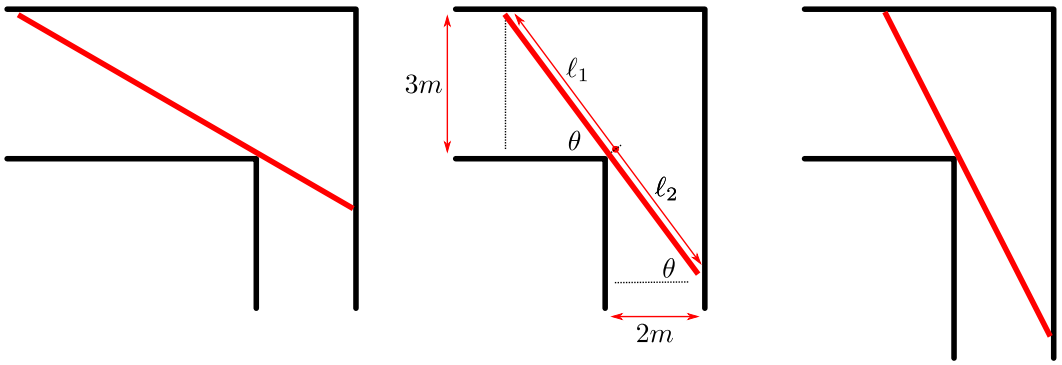

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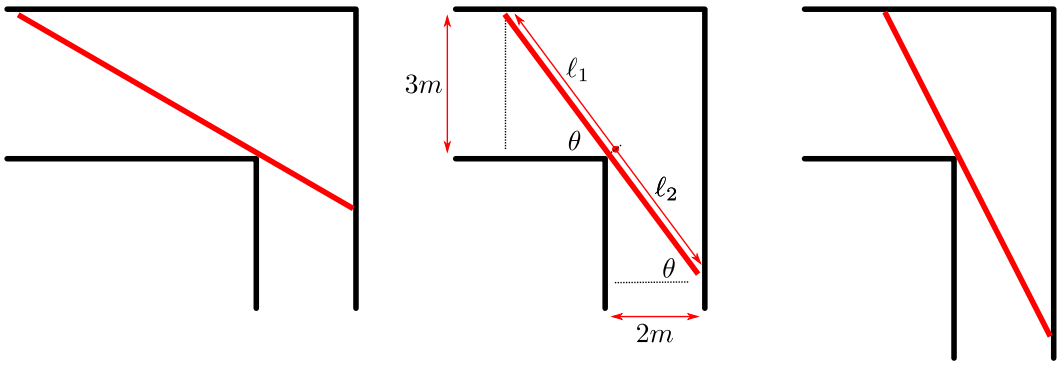

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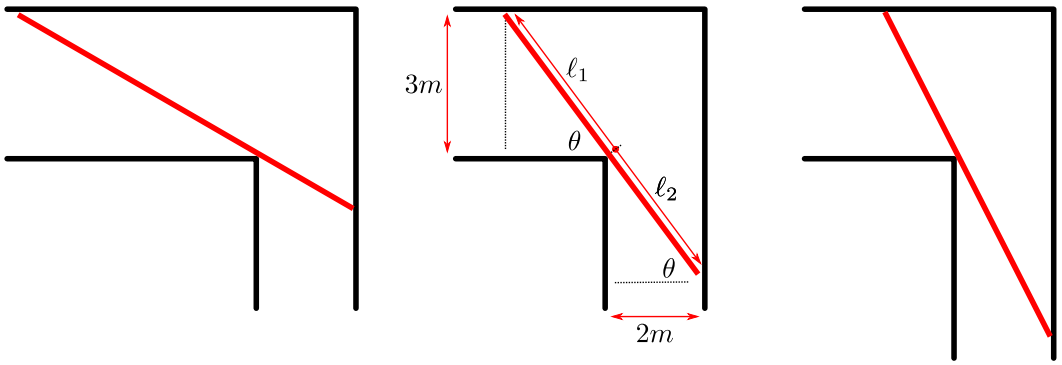

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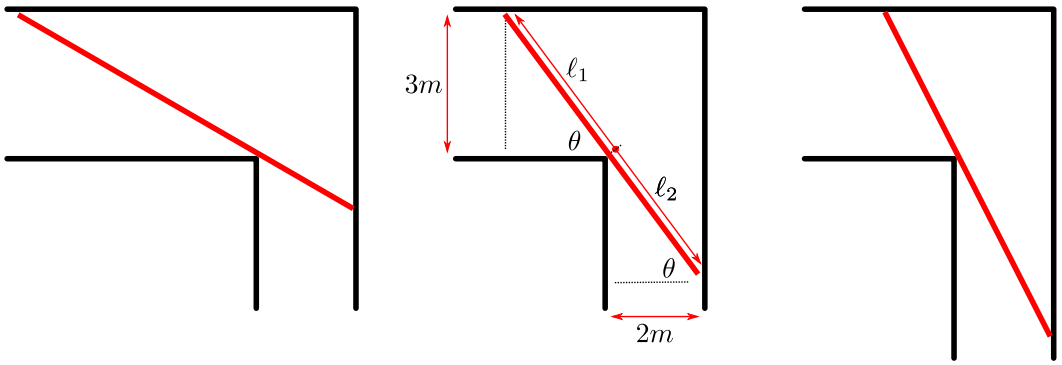

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Linear Approximations, Taylor Polynomials

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So $g'(c) = 0$ implies $f^{(n+1)}(c) - M = 0 \implies M = f^{(n+1)}(c)$.

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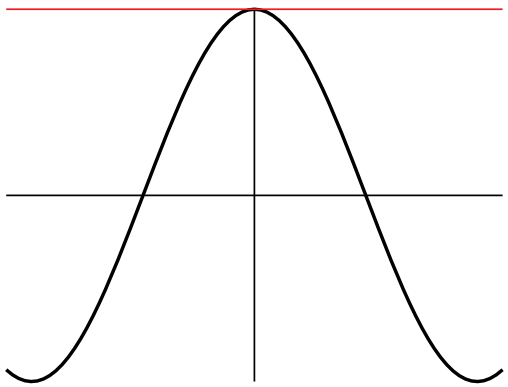
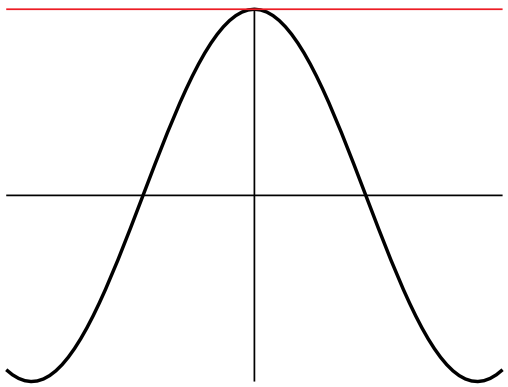


$\cos x \approx 1$

Maclaurin Polynomial Approximation of Cosine



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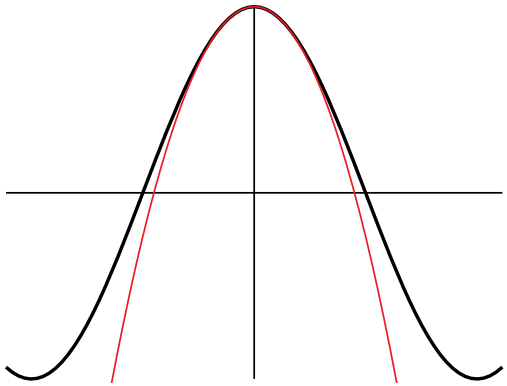

$\cos x \approx 1 - \frac{1}{2!}x^2$

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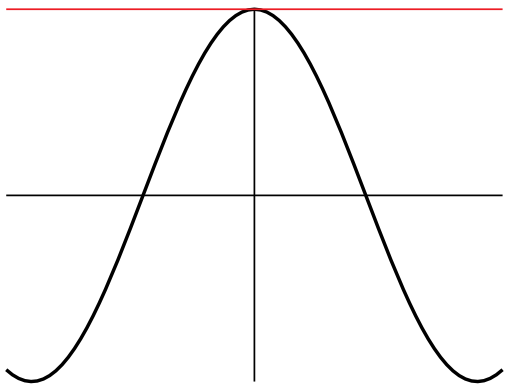

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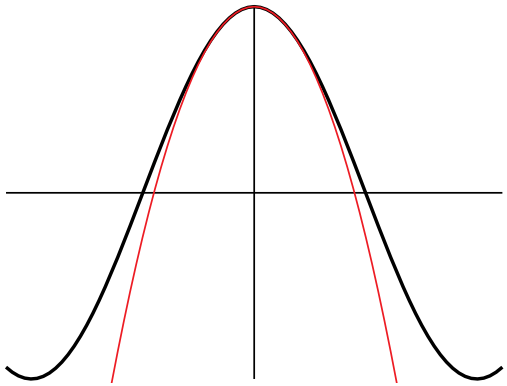

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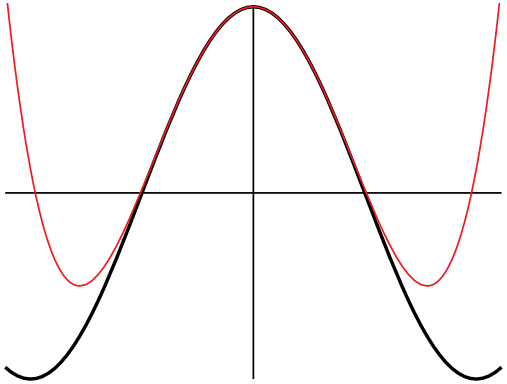

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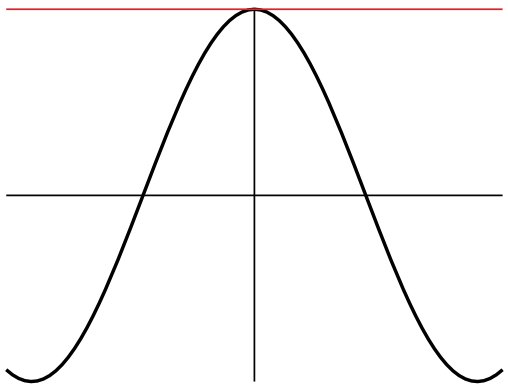

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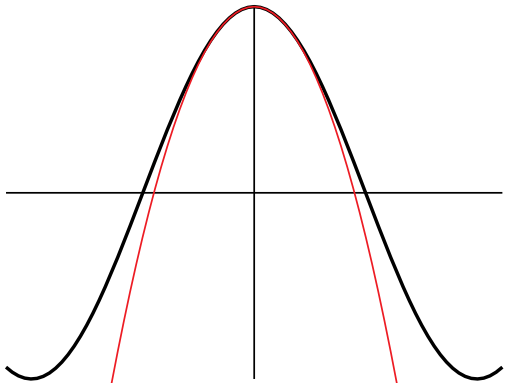

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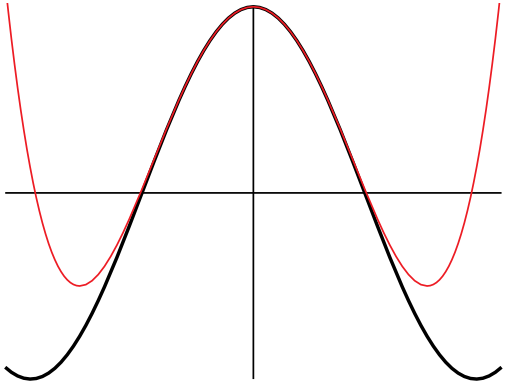

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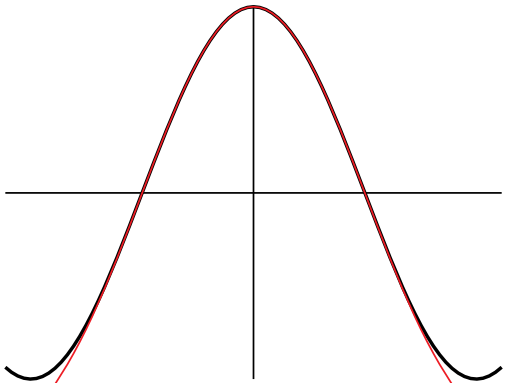

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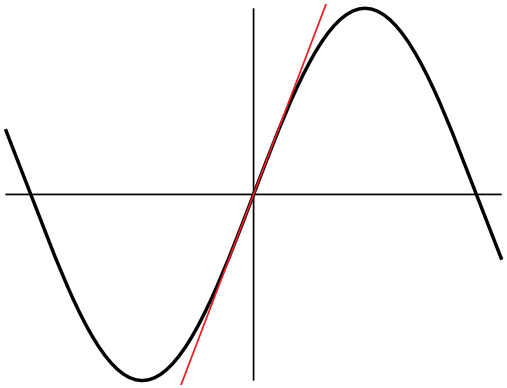

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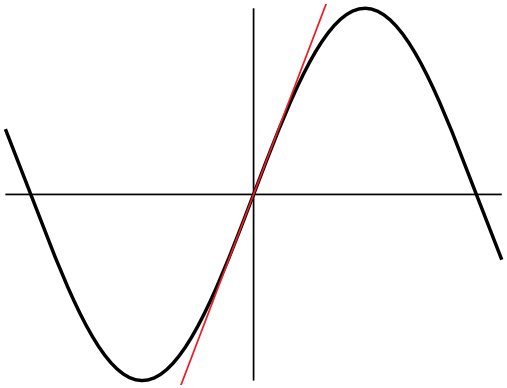
Maclaurin Polynomial Approximation of Sine



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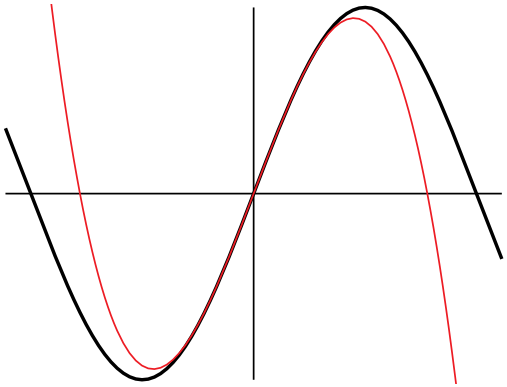

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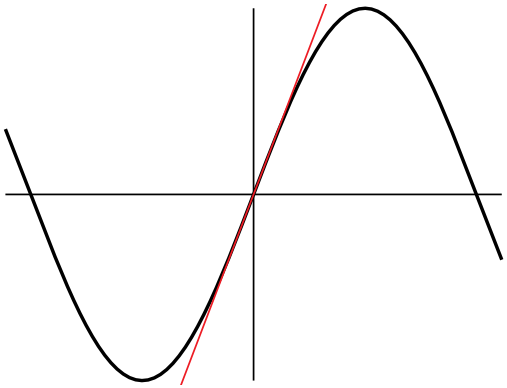

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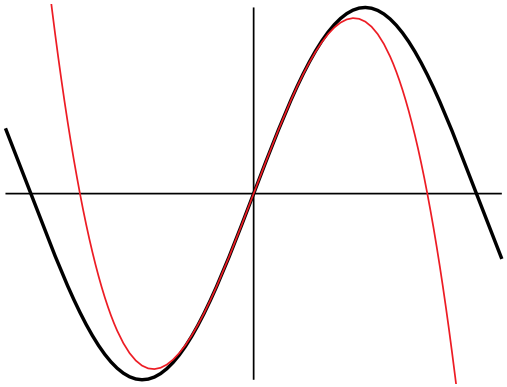
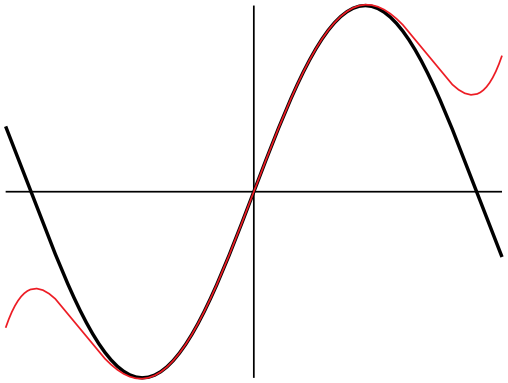
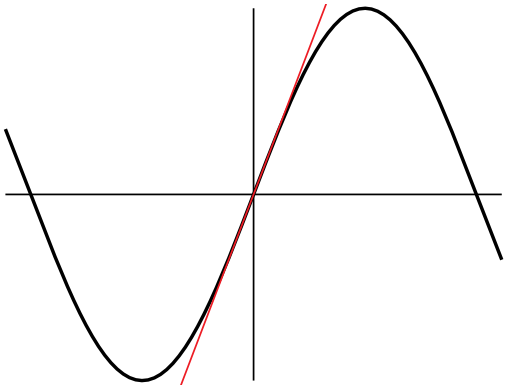


$\sin x \approx x - \frac{1}{3!}x^3$



$\sin x \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5$

Maclaurin Polynomial Approximation of Sine



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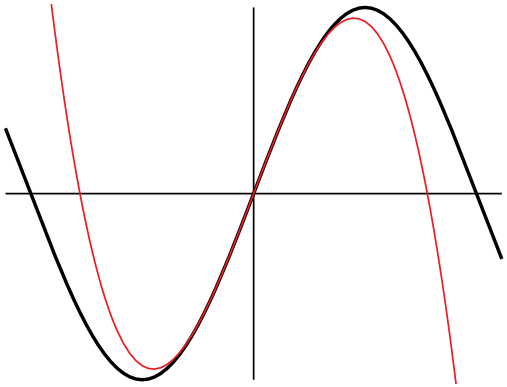

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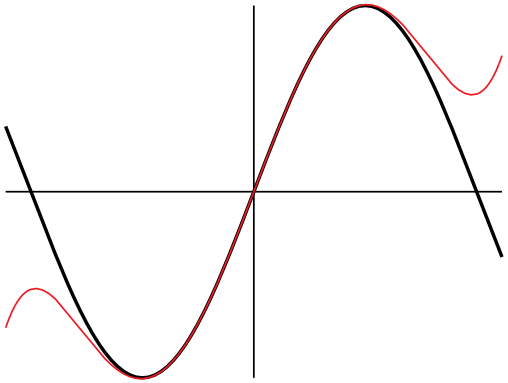

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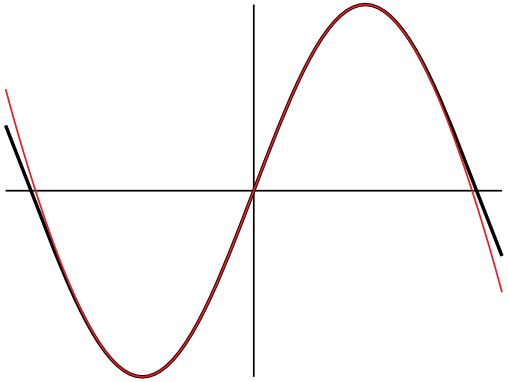
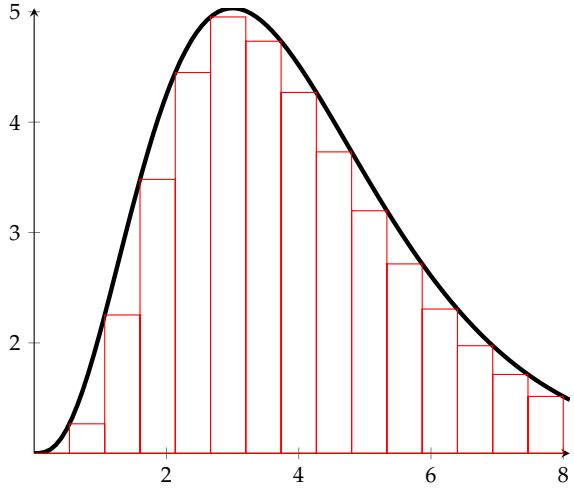


$\sin x \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7$

Ideas of Integration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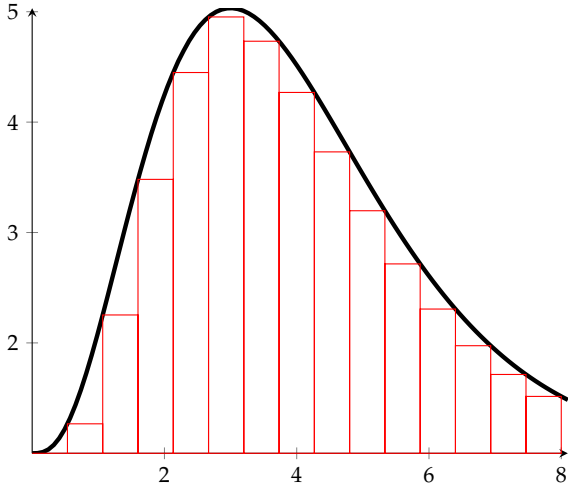
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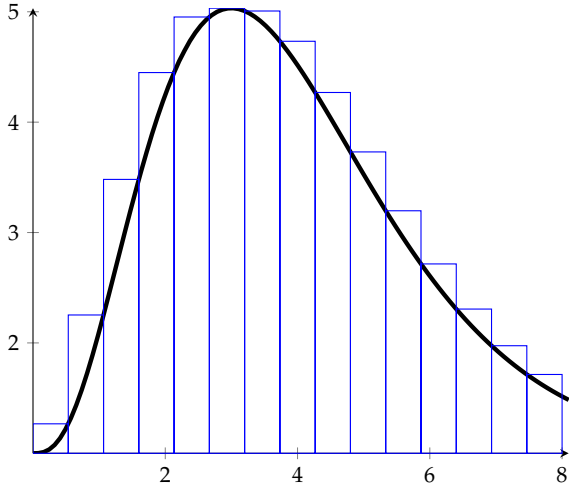
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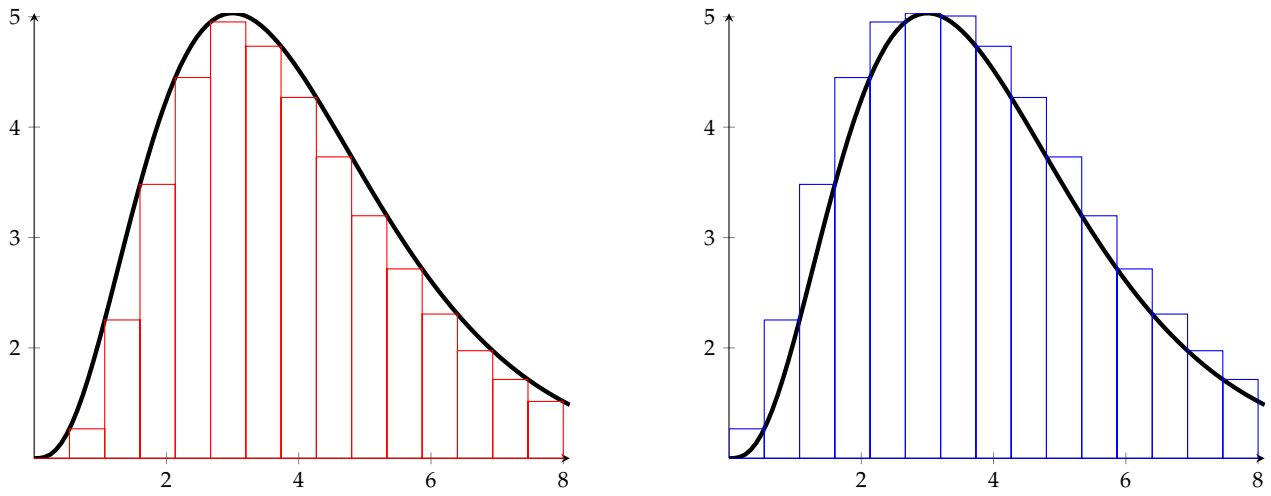

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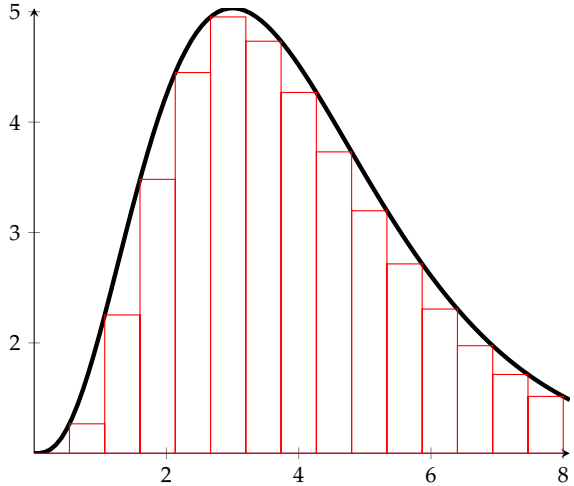

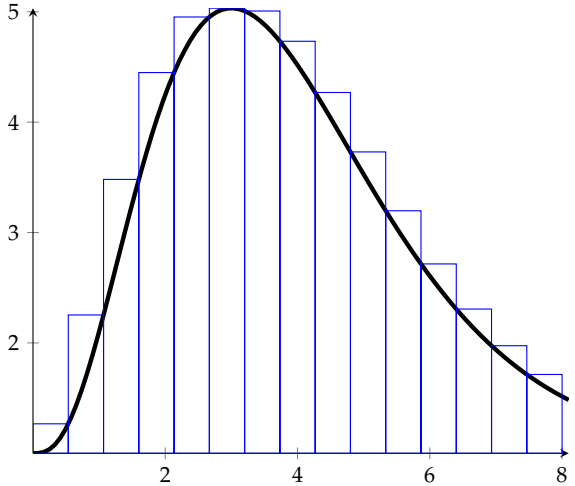

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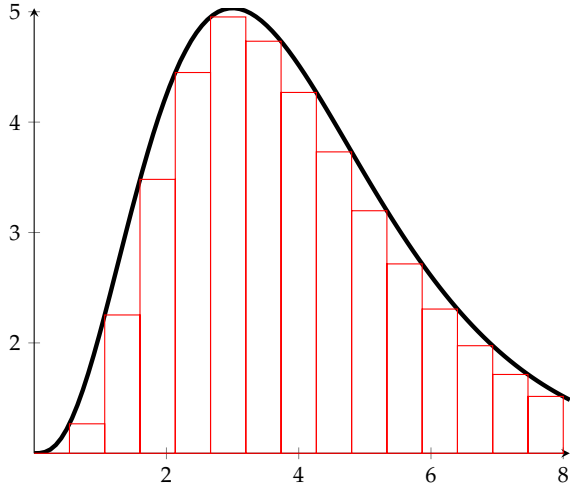

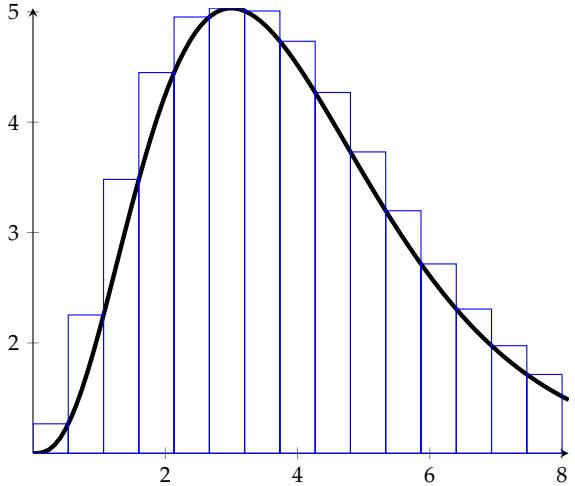

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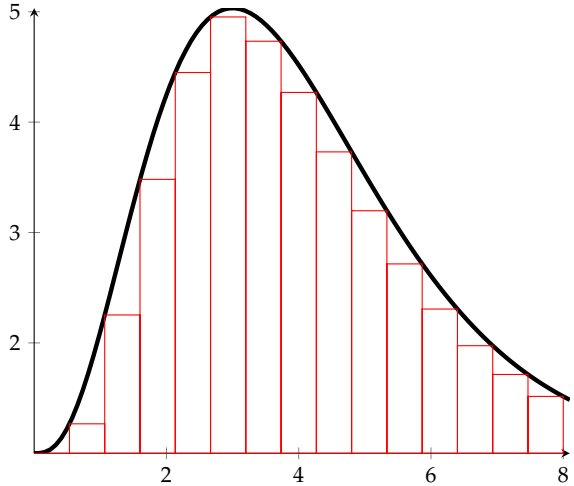

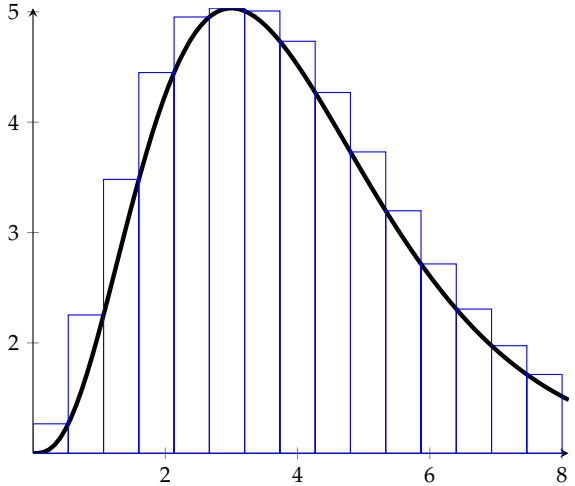

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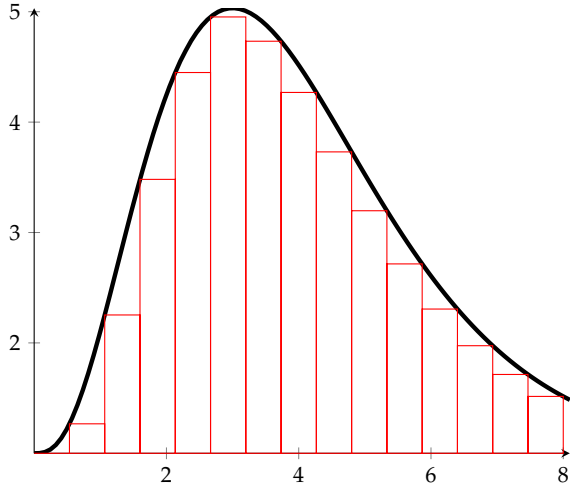

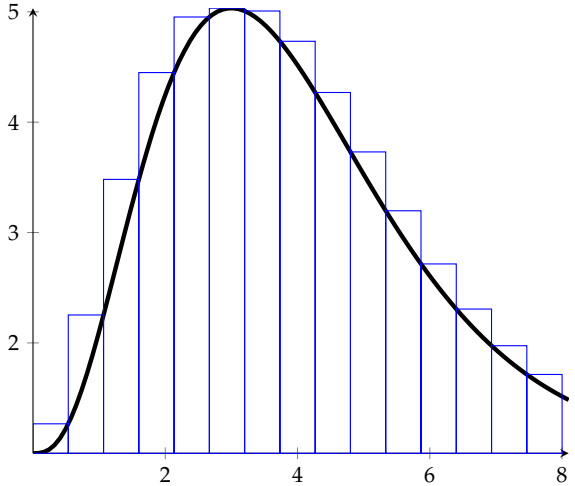

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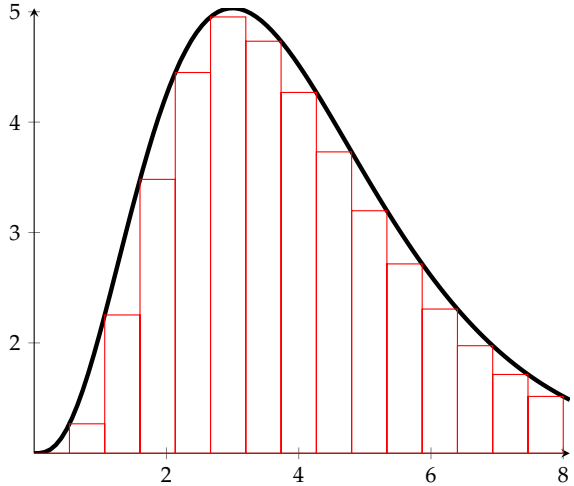
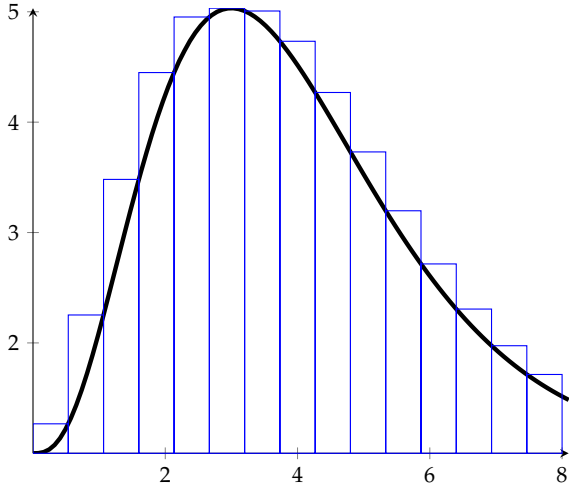
Define the sums $R(f, \mathbb{P}) = \sum_{k=1}^{n} f_k \Delta_k$,





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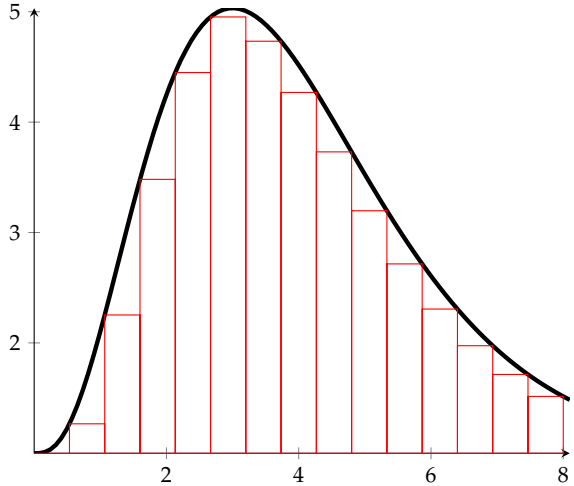

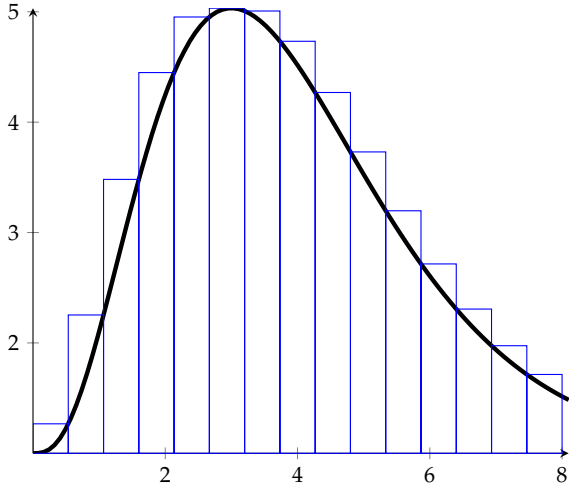

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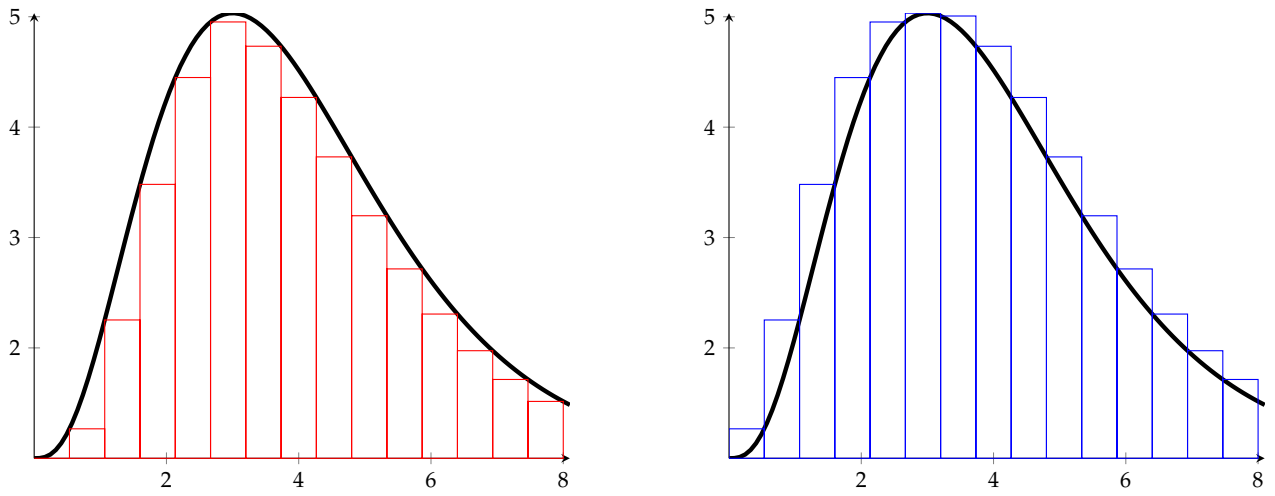

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A priori $L(f, \mathbb{P}) \leqslant R(f, \mathbb{P}) \leqslant U(f, \mathbb{P})$; Define

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\substack{n\to\infty \\ \|\mathbb{P}\|\to 0}} R(f, \mathbb{P})$$

Let f, g be integrable on an interval containing a, b, c, and constants α, β. Then

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Fundamental Theorem of Calculus (FTC): Example I

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The integral is $\lim_{n\to\infty} \sum_{k=1}^{n} \cos \frac{k\pi}{2n} \cdot \frac{\pi}{2n} = \lim_{n\to\infty} \frac{1}{2}\left(\frac{\sin\left(\left(n+\frac{1}{2}\right)\cdot\frac{\pi}{2n}\right)}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)} - 1\right) \cdot \frac{\pi}{2n} =$
$\lim_{n\to\infty} \frac{\sin\left(\left(n+\frac{1}{2}\right)\frac{\pi}{2n}\right)\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)} = \lim_{n\to\infty} \sin\left(\left(\frac{1}{2} + \frac{1}{4n}\right)\pi\right) \lim_{n\to\infty} \frac{\frac{1}{2}\cdot\frac{\pi}{2n}}{\sin\left(\frac{1}{2}\cdot\frac{\pi}{2n}\right)} = \sin\frac{\pi}{2} \cdot 1 = 1.$

The antiderivative of $\cos x$ is $\sin x$, for $(\sin x)' = \cos x$.

From FTC, $\int_0^{\frac{\pi}{2}} \cos x \, dx = \sin\frac{\pi}{2} - \sin 0 = 1$!

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$F'(x) = F_1'(x) + F_2'(x) = -x^3 + (2x)^3 \cdot 2 = 15x^3$.

Method of Substitution

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Elementary Integrals

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Method of Substitution: Integral Chain Rule

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Example: evaluate $\int \frac{x}{x^2+1}\,\mathrm{d}x$.

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Example: evaluate $\int \frac{\sin(3\ln x)}{x}\,\mathrm{d}x$.

From chain rule, $\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = f'(g(x)) \cdot g'(x)$, hence $\int f'(g(x)) \cdot g'(x)\,\mathrm{d}x = f(g(x)) + C$.

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